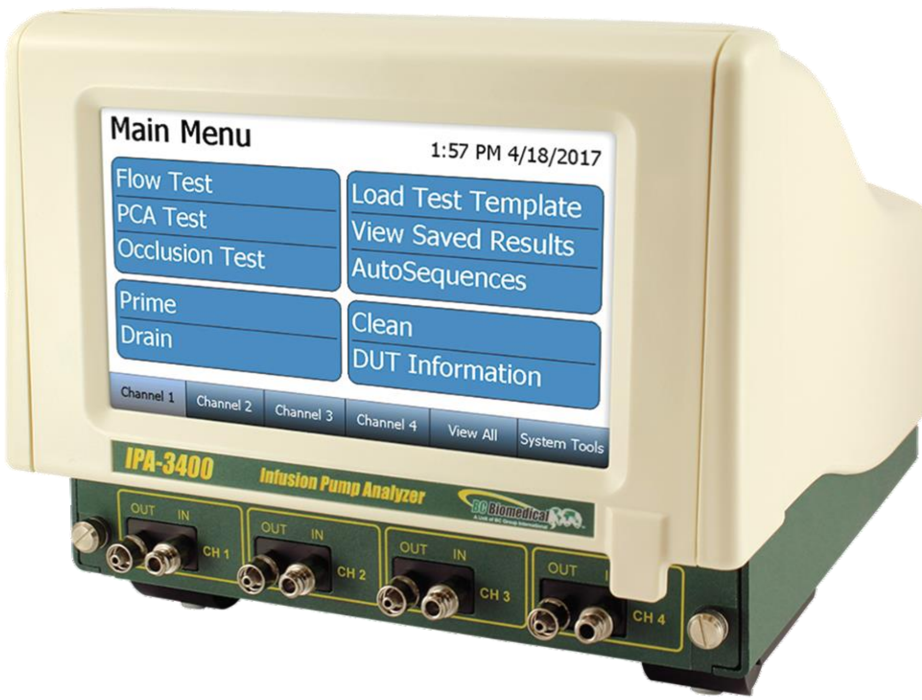
INFUSION PUMP ANALYZER



IPA-3400 SERIES

USER MANUAL

EU DECLARATION OF CONFORMITY

Manufacturer Name:	**BC Group International, Inc.**	**Tel:**	**+1 314 638 3800**
Manufacturer Address:	**3081 Elm Point Industrial Dr.**	**FAX:**	**+1 314 638 3200**
	St. Charles, MO 63301 USA	**EMAIL:**	sales@bcgroupintl.com

BC Group International, Inc. hereby declares under our sole responsibility that the product,

Product:	IPA-3400
Product Description:	Infusion Pump Analyzer
Year of CE marking:	2018

is in accordance with the European Union harmonized legislation as described within the following Directive(s):

2014/30/EU	ELECTROMAGNETIC COMPATIBILTY (EMC)
2014/35/EU	LOW VOLTAGE (LVD)
2011/65/EU	RESTRICTION OF THE USE OF CERTAIN HAZARDOUS SUBSTANCES IN ELECTRICAL AND ELECTRONIC EQUIPMENT (ROHS 2)

and in conformity with the following Harmonized Standard(s):

EN 61010-1:2010, 10th Edition	SAFETY REQUIREMENTS FOR ELECTRICAL EQUIPMENT FOR MEASUREMENT, CONTROL, AND LABORATORY USE – GENERAL REQUIREMENTS
EN 61326-1:2013, 13th Edition	ELECTRIACAL EQUIPMENT FOR MEASUREMENT, CONTROL AND LABORATORY USE – EMC REQUIREMENTS – PART 1: GENERAL REQUIREMENTS
EN 61000-3-2:2014, 2014 Edition	ELECTROMAGNETIC COMPATIBILITY (EMC) - PART 3-2: LIMITS - LIMITS FOR HARMONIC CURRENT EMISSIONS (EQUIPMENT INPUT CURRENT LESS THAN OR EQUAL TO 16 A PER PHASE)
EN 61000-3-3:2013, 2013 Edition	ELECTROMAGNETIC COMPATIBILITY (EMC) - PART 3-3: LIMITS - LIMITATION OF VOLTAGE CHANGES, VOLTAGE FLUCTUATIONS AND FLICKER IN PUBLIC LOW-VOLTAGE SUPPLY SYSTEMS, FOR EQUIPMENT WITH RATED CURRENT LESS THAN OR EQUAL TO 16 A PER PHASE AND NOT SUBJECT TO CONDITIONAL CONNECTION

A Technical file is maintained at the above address and is available, in full or in part, when additional documentation is required to show compliance with the applicable Directive(s) and Regulations referenced above.

I, the undersigned, of BC Group International, Inc. do hereby declare that the product(s) specified above, in its intended use as described within the product's User Manual, is in compliance with the directive(s) and standard(s) referenced above.

Place: BC Group International Inc.
3081 Elm Point Industrial Dr.
St. Charles, MO 63301 USA

Signature:

Mel Roche
President

Date: 11/12/2019

Document No.: 65-7316120-10-00
Revision: 00

Form: 65-700071-10 (00)

EU DECLARATION OF CONFORMITY

Manufacturer Name: BC Group International, Inc.
Manufacturer Address: 3081 Elm Point Industrial Dr.
St. Charles, MO 63301 USA

Tel: +1 314 638 3800
FAX: +1 314 638 3200
EMAIL: sales@bcgroupintl.com

BC Group International, Inc. hereby declares under our sole responsibility that the product,

Product: IPA-3900-FM
Product Description: Infusion Pump Analyzer Flow Module
Year of CE marking: 2018

is in accordance with the European Union harmonized legislation as described within the following Directive(s):

2014/30/EU	ELECTROMAGNETIC COMPATIBILTY (EMC)
2014/35/EU	LOW VOLTAGE (LVD)
2011/65/EU	RESTRICTION OF THE USE OF CERTAIN HAZARDOUS SUBSTANCES IN ELECTRICAL AND ELECTRONIC EQUIPMENT (ROHS 2)

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Place: BC Group International Inc.
3081 Elm Point Industrial Dr.
St. Charles, MO 63301 USA

Signature:

Mel Roche
President

Date: 11/12/2019

Document No.: 65-731979-10-00
Revision: 00

Form: 65-700071-10 (00)

BC BIOMEDICAL
IPA-3400 SERIES
TABLE OF CONTENTS

WARNING - USERS

The IPA-3400 Series is for use by skilled technical personnel only.

WARNING - USE

The IPA-3400 Series is intended for testing only and should never be used in diagnostics, treatment, or any other capacity where they would come in contact with a patient.

WARNING - CONNECTIONS

All connections to patients must be removed before connecting the DUT to the IPA-3400 Series. A serious hazard may occur if the patient is connected when testing with the unit. Do not connect any leads from the patient directly to the unit or DUT.

WARNING - LIQUIDS

Do not submerge or spill liquids on the IPA-3400 Series. Do not operate the IPA-3400 Series if internal components not intended for use with fluids may have been exposed to fluid, as the internal leakage may have caused corrosion and be a potential hazard.

WARNING - MODIFICATIONS

The IPA-3400 Series is intended for use within the published specifications. Any application beyond these specifications or any unauthorized user modifications may result in hazards or improper operation.

CAUTION - FLUID

Only Distilled or Deionized Water should be used in the chambers with the IPA-3400 Series. Do not use glucose, saline, or any other fluid; This will cause the fluid path to become contaminated. Flush with Distilled or Deionized water if fluid channels have been exposed to incompatible fluids – do not let the unit dry out with incompatible fluid in the Flow Module as the dried contaminants can buildup, leading to blockage.

CAUTION - SERVICE

The IPA-3400 Series is intended to be serviced only by authorized service personnel. Troubleshooting and service procedures should only be performed by qualified technical personnel.

CAUTION - ENVIRONMENT

Exposure to environmental conditions outside the specifications can adversely affect the performance of the IPA-3400 Series. Allow the IPA-3400 Series to acclimate to specified conditions for at least 30 minutes before attempting to operate it.

CAUTION - CLEANING

Do not immerse. The IPA-3400 Series should be cleaned by wiping gently with a damp, lint-free cloth. A mild detergent can be used if desired.

CAUTION - INSPECTION

The IPA-3400 Series should be inspected before each use for wear and should be serviced if any parts are in question.

NOTICE – SYMBOLS

Symbol	Description
	**Caution** (Consult Manual for Further Information)
	Per European Council Directive 2002/95/EC, do not dispose of this product as unsorted municipal waste.

NOTICE – ABBREVIATIONS

ANSI	**American National Standards Institute**
C	**Celsius**
°	**degree(s)**
DUT	**Device Under Test**
Euro	**European**
F	**Fahrenheit**
FS	**Full Scale**
hr	**hour(s)**
Hz	**hertz**
kg	**kilogram(s)**
kPa	**kilopascal(s)**
L	**liter(s)**
µL	**microliter(s)**
lbs	**Pounds**
mA	**milliampere(s)**
mBar	**milliBar(s)**
mL	**milliliter(s)**
mm	**millimeter(s)**
mmhg	**millimeters of mercury**
min	**minute(s)**
PSI	**Pounds per Square Inch Pressure**
RH	**Relative Humidity**
USA	**United States of America**
VAC	**Volts Alternating Current**

NOTICE – DISCLAIMER

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NOTICE – CONTACT INFORMATION

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BC BIOMEDICAL
IPA-3400 SERIES
INFUSION PUMP ANALYZER

The IPA-3400 is the most compact, full-featured, four-Channel analyzer on the market. It is a high-accuracy, easy-to-use system that incorporates touchscreen control of all processes without the use of old-fashioned buttons and knobs. This new, cutting-edge, patented design (US Patent No. 10,100,828) uses a dual syringe stepper motor-driven system that provides continuous monitoring of the fluid flow without the need to stop and perform intermittent drains as needed with older technologies. This provides a more realistic flow path for the Infusion Device under test and therefore more accurate readings. Independent stepper motor control of the custom ceramic valve allows the system to run smoothly and quietly.

The IPA-3400 has built-in AutoSequence capabilities that allows the user to perform automatic test procedures. This allows specific test routines specified by various manufacturers to be performed, providing significant time savings and reducing the risk of human error. All AutoSequence test results are stored internally in the large 32 GB memory. They can also be downloaded to a USB flash drive.

There are specific requirements in IEC 60601-2-24 for, not only flow readings, but also back pressure simulation, bolus (PCA) measurements, and occlusion alarm monitoring. All of these features are specifically built into the IPA-3400 with simple-to-use, onscreen selections.

The IPA-3400 is designed to hold up to four IPA-3900-FM Flow Modules. These Modules are individually serialized and calibrated so that they may be moved from Channel to Channel, and even unit to unit. Once installed, they are recognized by the IPA-3400 and their Serial Number and NIST Traceable Calibration information are presented on the display and utilized in all data reporting.

The interchangeable Modules provide the user with unprecedented flexibility in their IPA Testing System. There is no need to halt testing when the unit is due for calibration, simply replace the Module that is due with another Flow Module. Only the Modules need to be calibrated. This also allows the rotating of Modules or the use of a spare Module, thus providing flexibility not available in other systems. The same is true for service. If there is a problem on one Channel, only the Module needs to be serviced, eliminating downtime.

The user has complete access to the four interchangeable Flow Modules. No wiring or plumbing is required for Module installation. Simply lift the IPA-3400 display, remove the Access Panel, loosen the Module’s retainer screw, and slide the Module out. All plumbing and electronics are self-contained, and all electrical connections are automatically mated. Simply reverse the process for Module installation.

The IPA-3400 easily allows expansion in the field. You can buy a one-Channel unit and purchase additional Modules to be installed in the field at any time. Just plug them in and upon startup the system will automatically recognize the additional Module(s) and reconfigure itself accordingly.

IPA-3400 SERIES (BASIC FEATURES):

- SMALL IN SIZE, LARGE IN FEATURES
- FAST OPERATION
- EASY TO USE
- HIGH-ACCURACY
- LARGE 7” COLOR TOUCHSCREEN DISPLAY
- 1, 2, 3, AND 4 CHANNEL MODELS AVAILABLE (FIELD-UPGRADEABLE)
- INTERCHANGEABLE, FULLY SELF-CONTAINED FLOW MODULES
- CALIBRATION STORED IN THE FLOW MODULES – NO NEED TO BE DOWN FOR CALIBRATION OR SERVICE
- SMOOTH DUAL SYRINGE SYSTEM – ELIMINATES DRAIN CYCLE INCONSISTENCIES
- WHISPER-QUIET OPERATION
- AUTO START TESTS
- BUILT-IN DATA COLLECTION
- INDUSTRIAL GRADE SS PRESSURE SENSOR
- PERFORMS ALL IEC 60601-2-24 REQUIRED TESTS
- 100 ul to 1600 mL/Hr
- 4 USB PORTS, 4 AUX PORTS
- FLASH DRIVES, BARCODE SCANNERS, KEYBOARD AND MOUSE DIRECTLY SUPPORTED
- CONFIGUREABLE PRESSURE UNITS (mmHg, PSI, Bar, kPa)
- LARGE 32GB INTERNAL MEMORY

FUNCTIONS:

- PCA/BOLUS
- BACK PRESSURE SIMULATION
- OCCLUSION ALARM
- DATA DOWNLOAD TO FLASH DRIVE
- SELF-CLEANING CYCLE

AVAILABLE MODELS:

- IPA-3400-1 INFUSION PUMP ANALYZER BENCH TOP – MULTI-CHANNEL W/ 1 FLOW MODULE (IPA-3900-FM)
- IPA-3400-2 INFUSION PUMP ANALYZER BENCH TOP – MULTI-CHANNEL W/ 2 FLOW MODULES (IPA-3900-FM)
- IPA-3400-3 INFUSION PUMP ANALYZER BENCH TOP – MULTI-CHANNEL W/ 3 FLOW MODULES (IPA-3900-FM)
- IPA-3400-4 INFUSION PUMP ANALYZER BENCH TOP – MULTI-CHANNEL W/ 4 FLOW MODULES (IPA-3900-FM)

ACCESSORIES:

- BC20-00161 IPA-3400 ACCESSORY KIT
- BC20-40617 IPA-3400 TUBING KIT
- BC20-20521 POWER CORD WITH INTERCHANGEABLE US, UK, AUS, AND EURO PLUGS
- BC20-41360 CABLE, COMMUNICATIONS NULL MODEM
- BC20-41370 BC FLOW SOFTWARE CD
- IPA-3900-FM FLOW MODULE
- IPA-3900-MB MODULE BLANK

BC20-20521

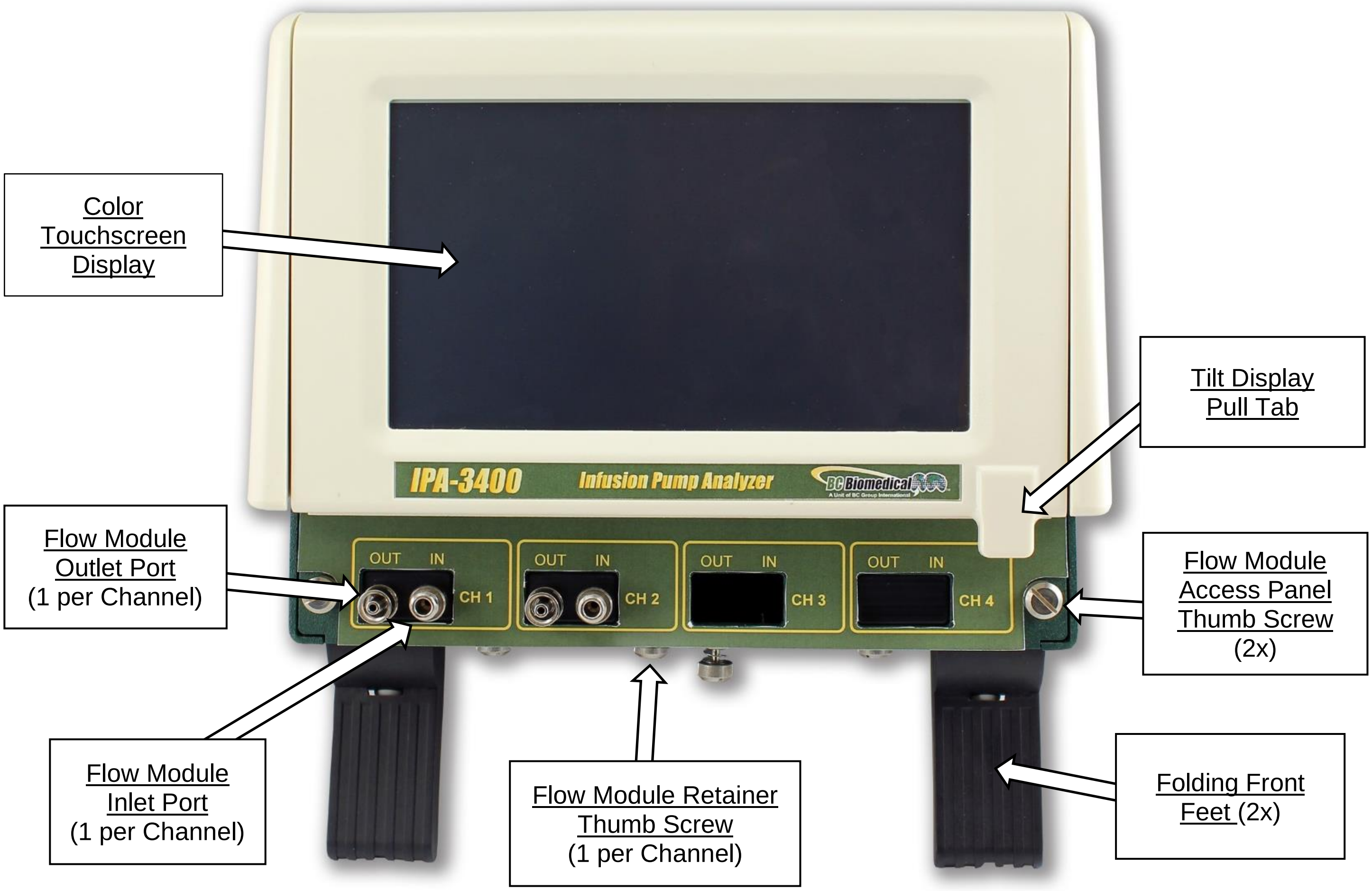
BC20-41360

BC20-40617

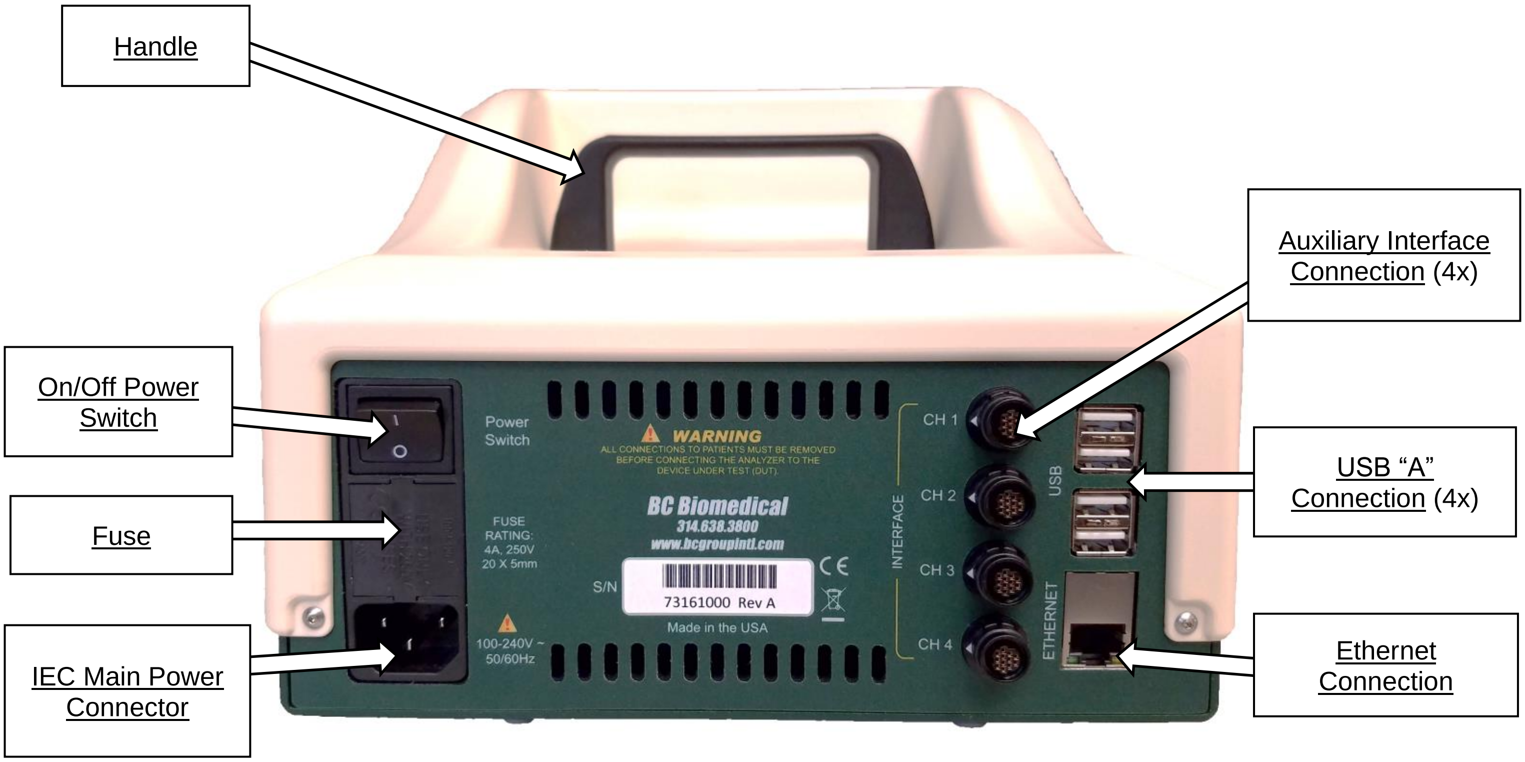
OVERVIEW

Below are details of the IPA-3400 Series layout and components:

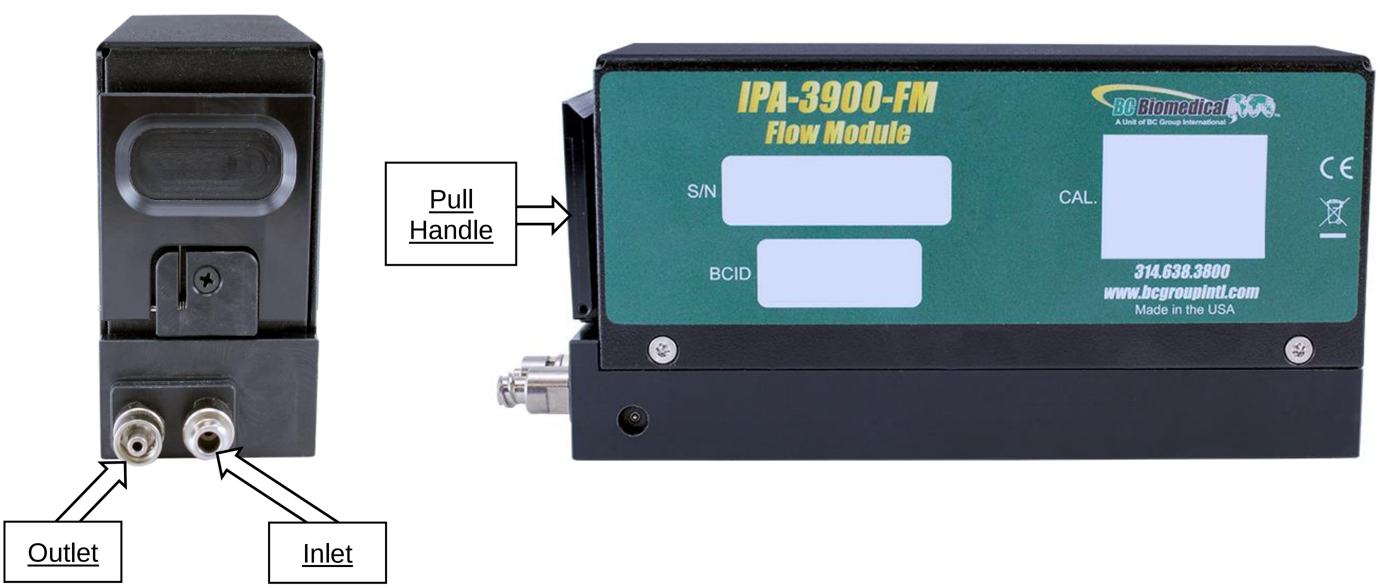
FRONT VIEW



BACK VIEW



IPA-3900-FM



GETTING STARTED

CONNECTIONS

The IPA-3400 allows for a variety of connections to meet specific customer and test requirements. Not all connections are necessary for each application, but all are listed for completeness. The order in which the connections are made is not important. The Module initialization on power up will draw a vacuum on the inlet port and could cause the Module to stall if not open to atmosphere.

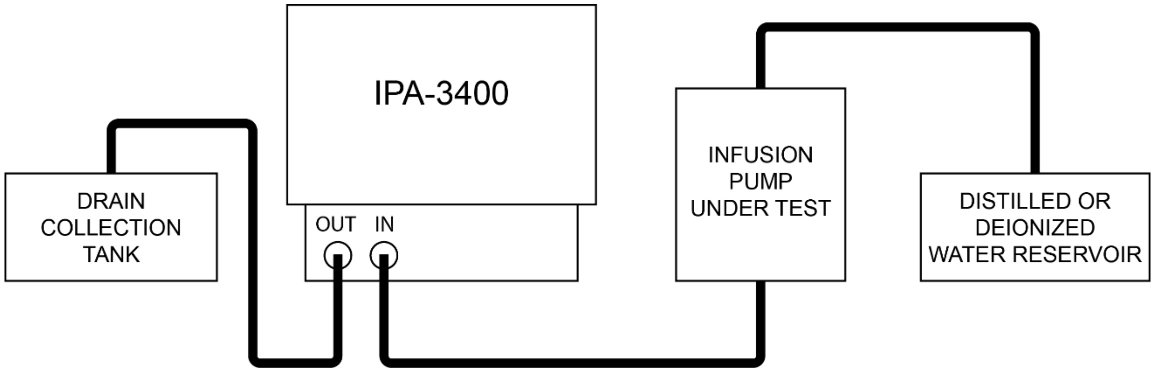
Power

Plug the power cord into the standard IEC connector on the IPA-3400 back panel. Select an included plug adapter that matches the required wall receptacle. Connect the adapter into the mating end of the power cord, and then into the wall receptacle.

Fluid connections

IPA-3400 utilizes an independent, interchangeable Flow Module for each Channel. These are intelligent Modules that contain all necessary Hardware and Software (including calibration data) to operate a single Channel of fluid analysis. The Modules may be placed in any of the four available Channel slots. The IPA-3400 automatically identifies and communicates with up to four installed Modules on startup.

All Modules and their connections are identical, therefore only a single Module is detailed. The process may then be repeated for the additional Modules installed, for additional infusion pumps under test. Each Module has an inlet and an outlet Port. These utilize standard Luer connectors. The diagram below illustrates a typical connection.



It should be noted that the advanced design in the IPA-3400 is far less susceptible to the normal restrictions of older infusion pump testing technologies. In particular, the system is much more forgiving of tubing lengths and air bubbles. While the accuracy of the system can be affected by these issues, air bubbles will not impact the operation of the system and it is not necessary for the system be level to operate correctly.

Priming the Channels

Once the fluid connections have been made, the system should be primed. Use the onscreen Prime button for the Channel connected. The fluid source should be run until no air bubbles are seen in the outlet tube.

Accessory connections

Accessory connections are made to the appropriate connectors on the back panel.

USB - There are four USB "A" connectors on the back panel. Any of the connectors may be utilized for the connection of supported USB accessories (Keyboard, Mouse, Bar Code Reader, or Flash Drive).

Keyboard -	Any standard USB keyboard may be used. The keyboard is not necessary for normal operation, since there is a built-in keyboard in the touchscreen
Mouse -	Any standard wired or wireless mouse may be used. The mouse is not necessary to operate the system since the touchscreen is available for all operations. However, the mouse may also be used to control all functions, if desired.
Bar Code Reader -	Any standard Bar Code Reader may be used to scan in device information if desired.
Flash Drive -	Any standard FAT32 formatted USB flash drive may be used to both upload and download information. This may be used for test data files, setup files or system software updates.

Ethernet - A standard RJ-45 Ethernet connector is provided on the back panel. This may be used to connect the system to a network, allowing operation of the system over the network. This may also be used for the uploading and downloading of test data files, setup files, or system software updates.

RUNNING A TEST - General

Once the system is connected and primed you are ready to run a test.

There are three basic tests - **Flow, PCA,** and **Occlusion** .

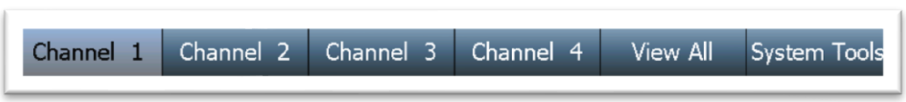
Each type of test may be performed individually, or as part of an **AutoSequence.**

Individual test setups may be created independently or loaded from a previously saved setup file.

AutoSequences are sets of consecutive tests with optional informational screens included. These are generated to follow specific sequences of tests that may follow OEM procedures or in-house custom procedures.

General Navigation Keys

There are 6 keys across the bottom of the screen that allow for simple navigation to the desired function.

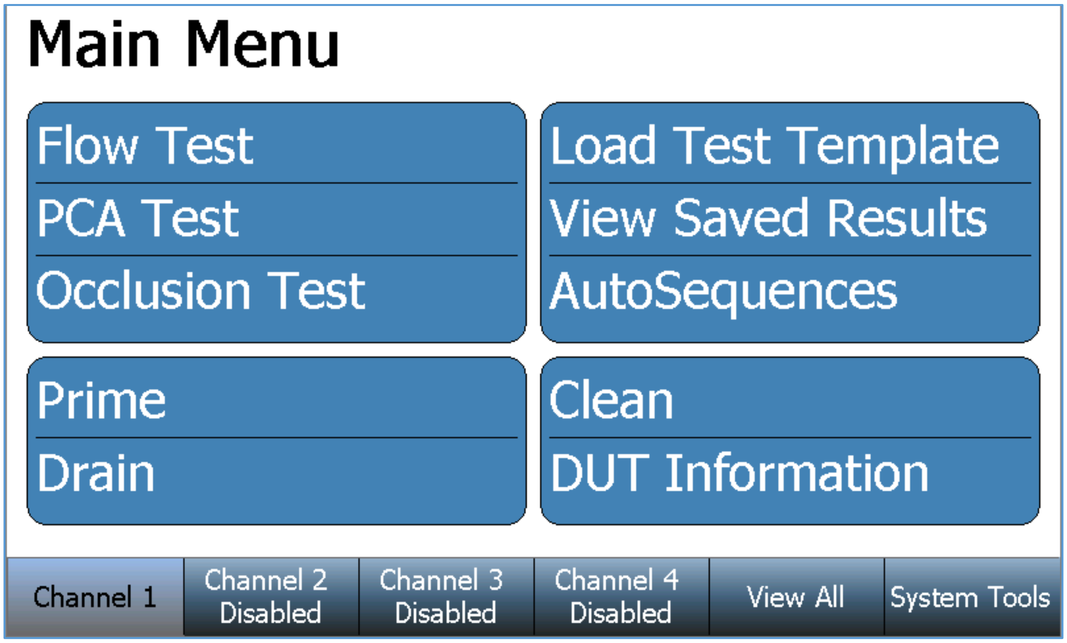


Channels 1 thru 4 - Pressing one of these keys displays the selected Channel functions and includes specific information about the Channel setup or test progress. These keys are only accessible when a Module is installed in the corresponding Channel slot.

View All - Pressing this key displays a single screen view containing the status of each installed Channel.

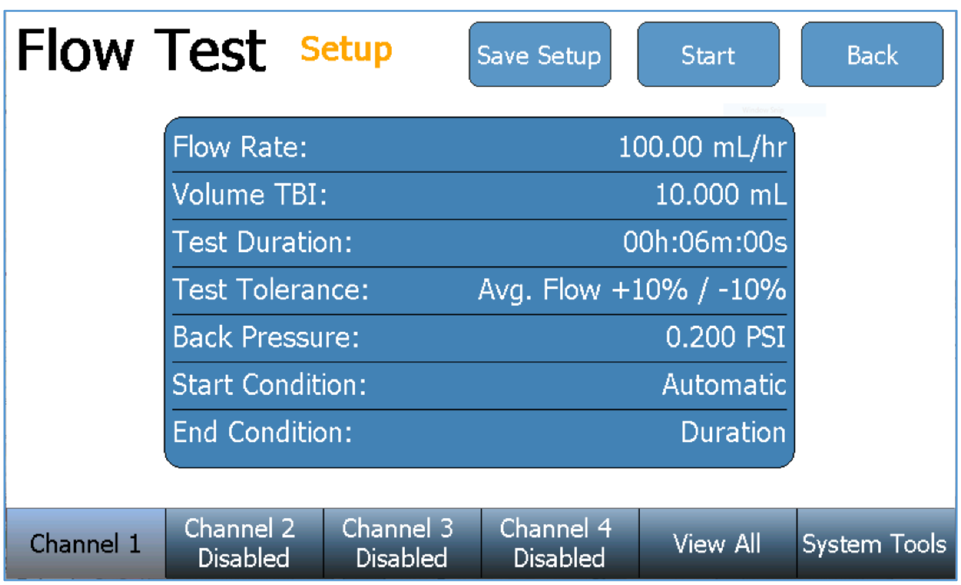
System Tools - Pressing this key opens the System Tools Screen, which provides access to the various utilities and system setup parameters.

RUNNING A FLOW TEST



Above is the Main Menu screen for a selected Channel. If needed, press the **Back** key on any successive screen to return to the Channel's Main Menu. All installed Channels have identical screen features and functions.

To setup a flow test, press the **Flow Test** key from the selected Channel's Main Menu. The following Flow Test setup screen will appear:

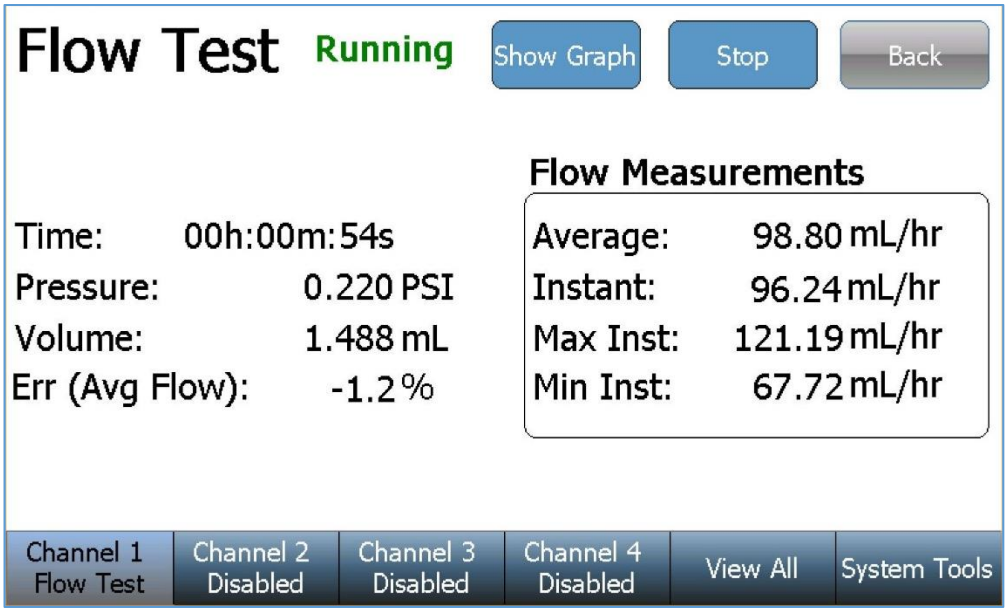


The Flow Test setup screen includes the flow test setup and control parameters. To change a setting, simply press the parameter to display the available options, select an option, or enter the desired value. Use the **Save Setup** key to save to save the current selections for use with another Channel or another IPA-3400. See the “Loading a Test Template” section of this manual for details.

Parameter	Range	Units
Flow Rate	0 to 1600	mL/hr
Volume To Be Infused	0 to 9999	mL
Test Duration	119:59:59	h:m:s
Test Tolerance:	------	------
(Test Parameter)	Volume, Instantaneous Flow, Average Flow, None	------
(Test Tolerance - %)	0 - 100	%
(Test Tolerance - Value)	Volume (0 to 9999) Flow (0 to 1600)	mL mL/hr
Back Pressure	-3.867 to 11.602	PSI
Start Condition*	Automatic or Manual	------
End Condition	Duration, Volume, Duration or Volume, Duration and Volume	------

* Automatic start indicates that a test will begin when the **Start** key is pressed and flow is detected.
Manual start indicates that the test will start immediately when the **Start** key is pressed.

Once the desired conditions have been set, press the **Start** key to initiate the Flow Test. The screen will change to the following:

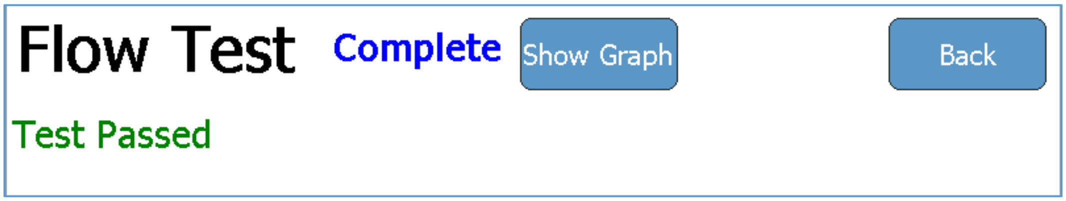


The **Show Graph** key may be used at any time during or after the test to change the screen to the Graph mode as shown below:

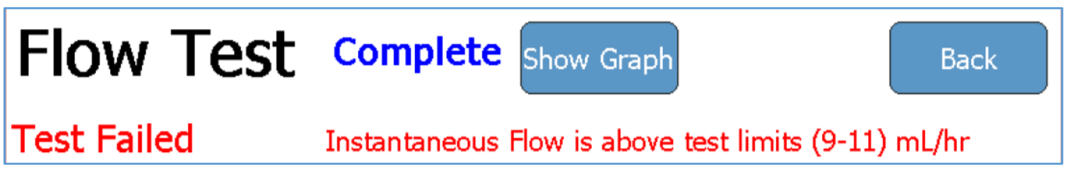
In the Graph mode, several different views of the data may be displayed. Use the keys on the right to select the desired display. Use the **Hide Graph** key to return to the previous display.

The graph can display Average Flow, Instant Flow, Average & Instant Flow, Volume, or Percent Error. The Avg Flow button toggles between Average Flow, Instant Flow, and Average & Instant Flow.

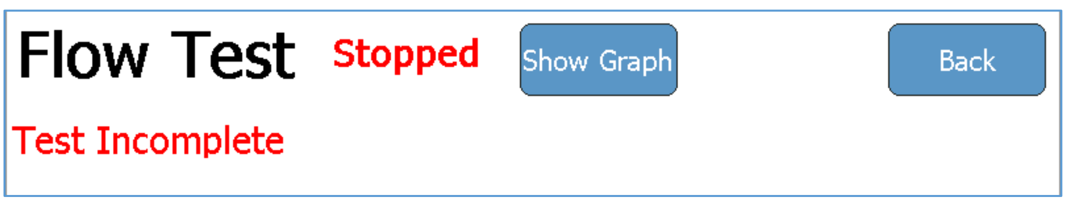
The test will continue until completion, unless manually stopped. Upon test completion, the test status will change from **Running** to **Complete** and the pass/fail test result will be displayed:



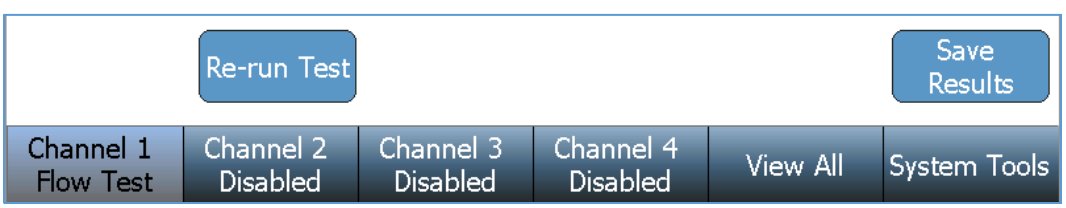
If the test has failed, a brief description of the failure is included:



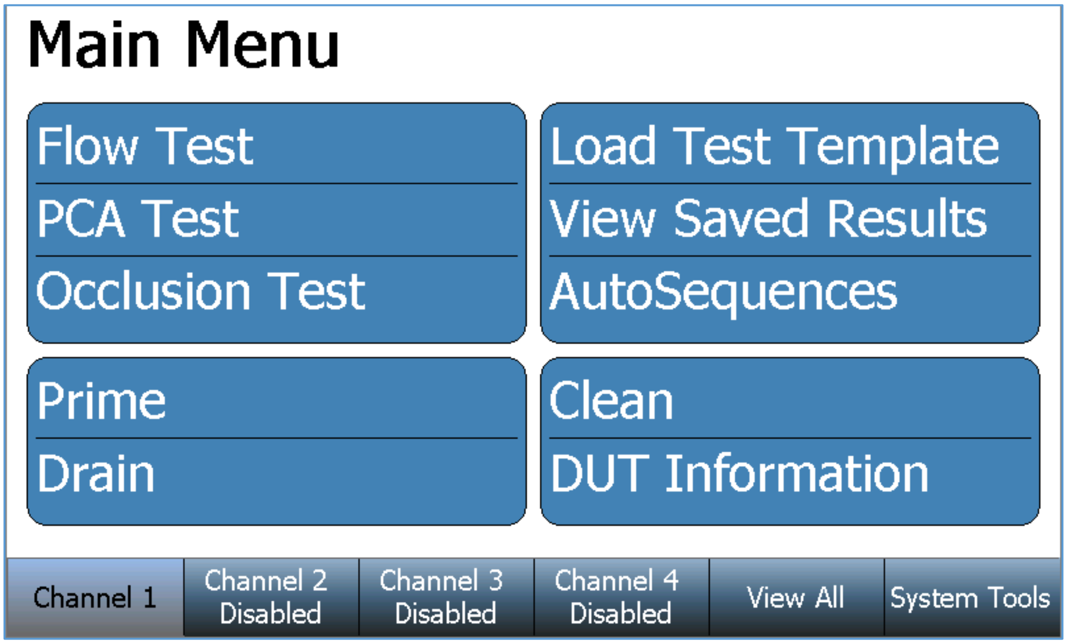
If the test is manually stopped, the test result will change from **Running** to **Stopped** and the test result will indicate an incomplete test:



Once a test has completed (or stopped) a **Re-run Test** key and a **Save Results** key will appear near the bottom of the screen. The test may be repeated using the **Re-run Test** key or the test results may be saved using the **Save Results** key:

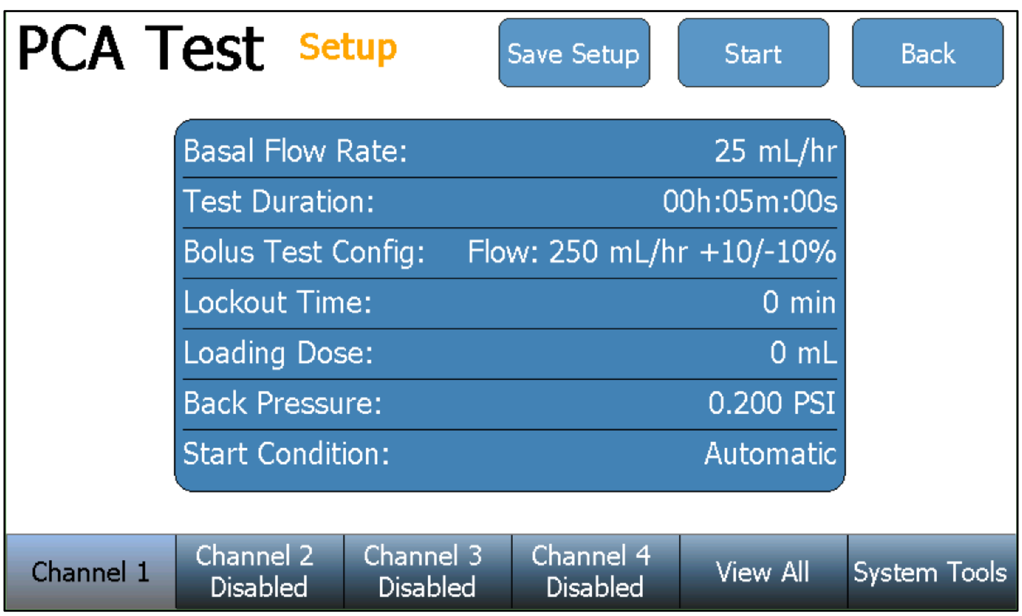


RUNNING A PCA TEST



Above is the Main Menu screen for a selected Channel. If needed, press the **Back** key on any successive screen to return to the Channel's Main Menu. All installed Channels have identical screen features and functions.

To setup a PCA Test, Press the **PCA Test** key from the selected Channel's Main Menu. The following PCA Test setup screen will appear:



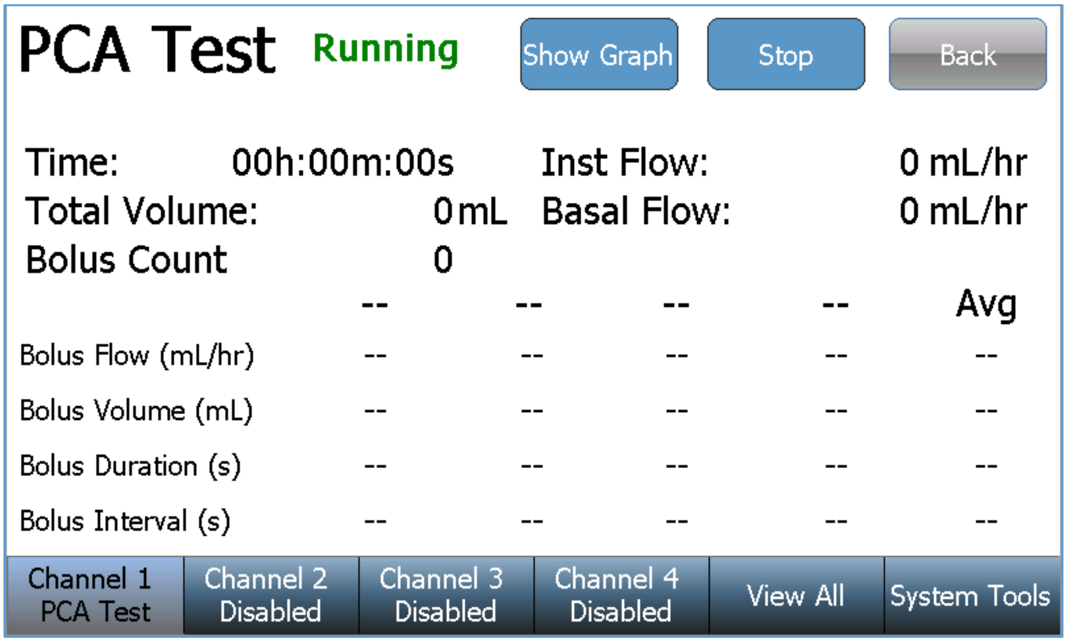
The PCA Test setup screen includes the PCA test setup and control parameters. To change a setting, simply press the parameter to display the available options, select an option, or enter the desired value. Use the **Save Setup** key to save to save the current selections for use with another Channel or another IPA-3400. See the “Loading a Test Template” section of this manual for details.

Parameter	Range	Units
Basal Flow Rate	0 to 1600	mL/hr
Test Duration	119:59:59	h:m:s
Bolus Test Config:	------	------
(Test Parameter):	Volume, Flow, None	------
(Expected Value)	Volume (0 to 9999) Flow (0 to 1600)	mL mL/hr
(Test Tolerance - %)	Upper (0 to 100) Lower (0 to 100)	%
(Test Tolerance - Value)	Upper (0 to 1600) Lower (0 to 1600)	mL/hr
Lockout Time*	0 to 999	minutes
Loading Dose*	0 to 9999	mL
Back Pressure	-3.867 to 11.602	PSI
Start Condition**	Automatic or Manual	------

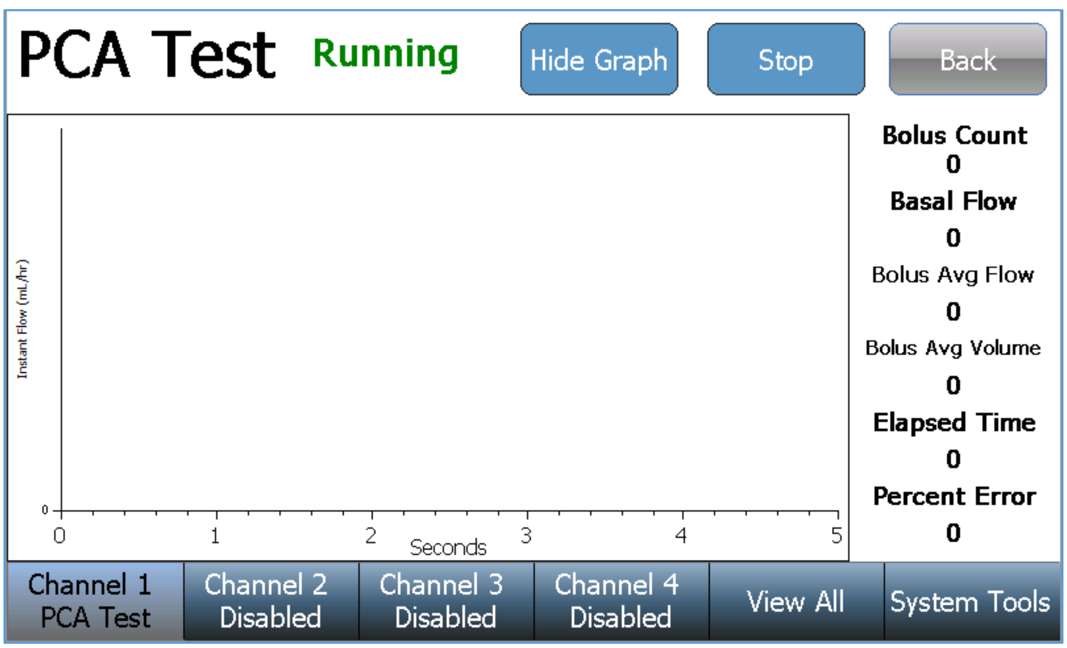
* For Report use only

** Automatic start indicates that a test will begin when the **Start** key is pressed and flow is detected. Manual start indicates that the test will start immediately when the **Start** key is pressed.

Once the desired conditions have been set, press the **Start** key to initiate the PCA Test. The screen will change to the following:

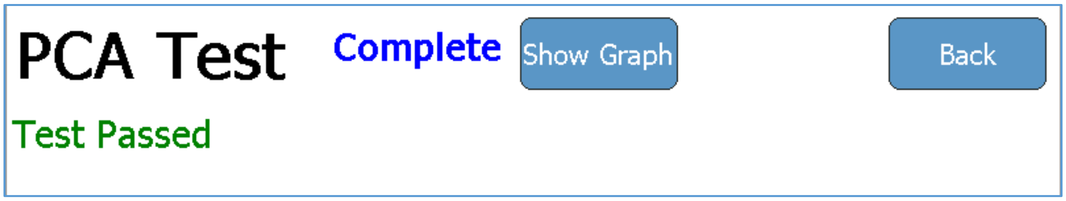


The **Show Graph** key may be used at any time during or after the test to change the screen to the Graph mode as shown below:

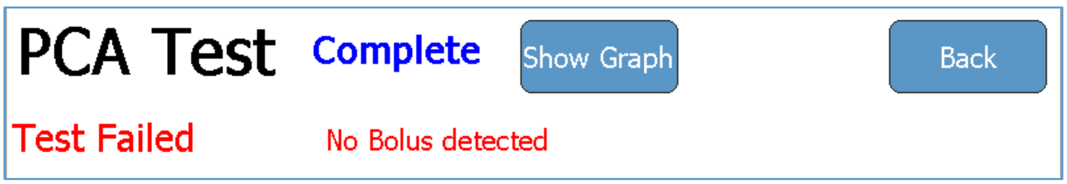


Use the **Hide Graph** key to return to the previous display.

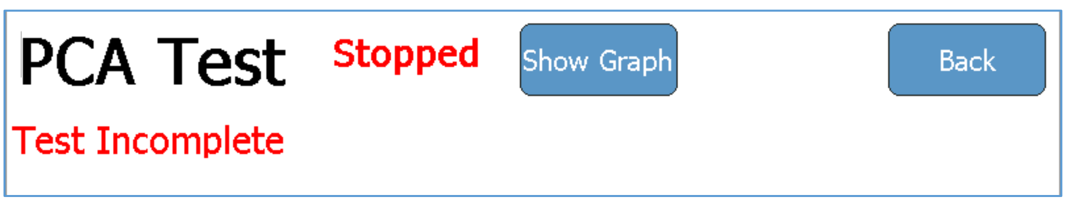
The test will continue until completion unless manually stopped. Upon test completion, the test status will change from **Running** to **Complete** and the pass/fail test result will be displayed:



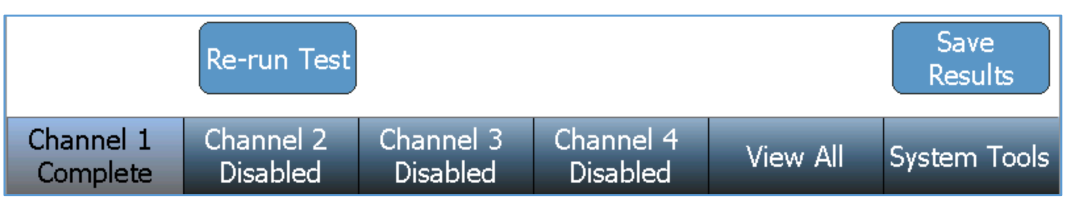
If the test has failed, a brief description of the failure is included:



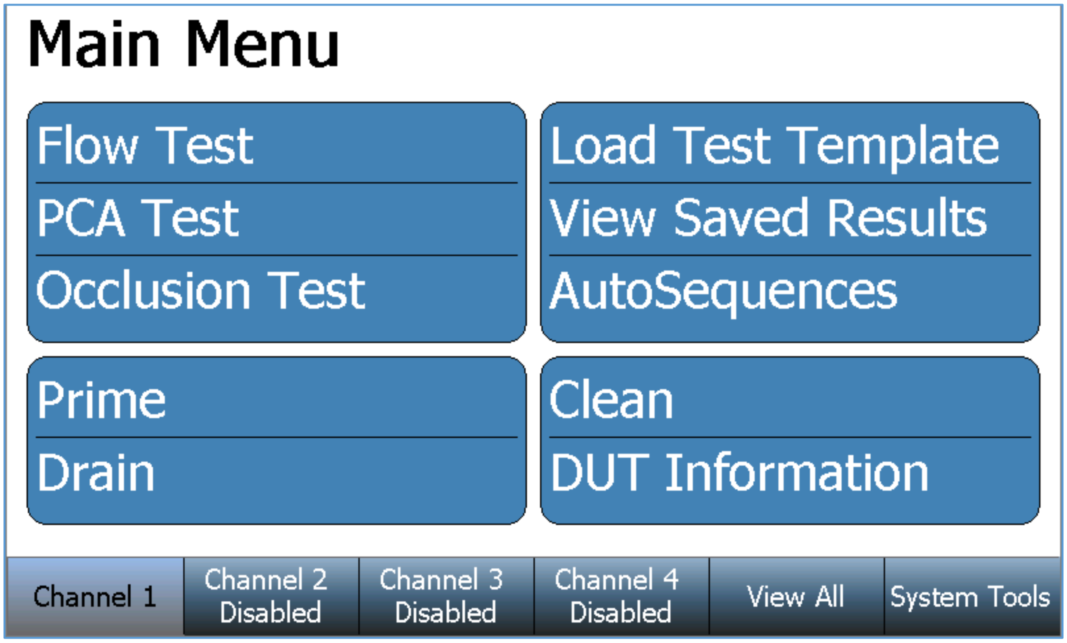
If the test is manually stopped, the test result will change from **Running** to **Stopped** and the test result will indicate an incomplete test:



Once a test has completed (or stopped) a **Re-run Test** key and a **Save Results** key will appear near the bottom of the screen. The test may be repeated using the **Re-run Test** key or the test results may be saved using the **Save Results** key:

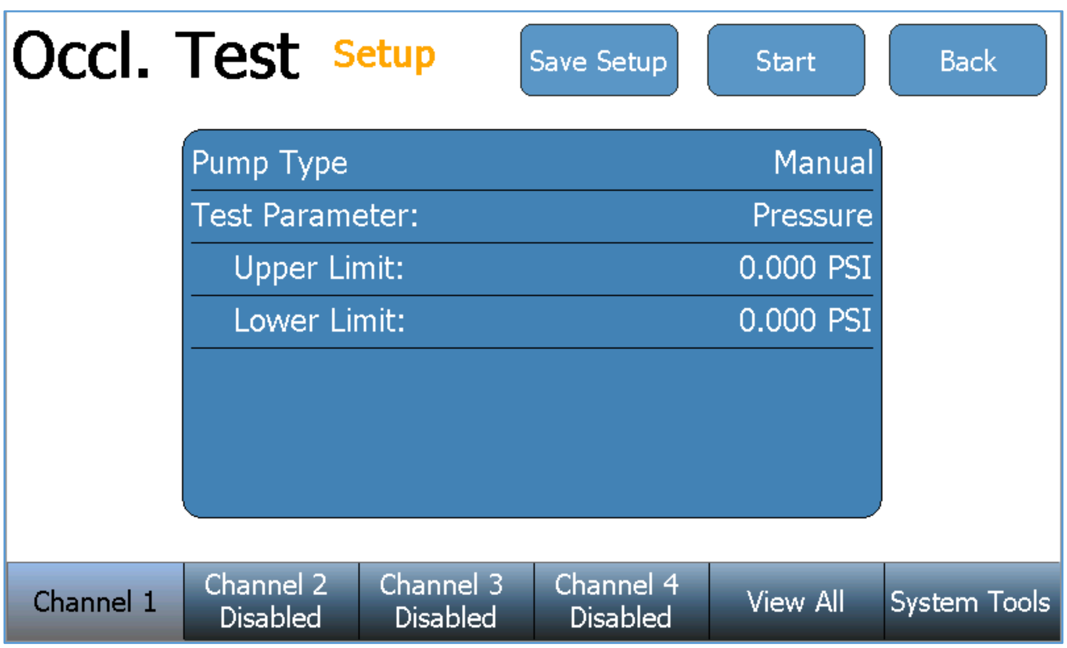


RUNNING AN OCCLUSION TEST



Above is the Main Menu screen for a selected Channel. If needed, press the **Back** key on any successive screen to return to the Channel's Main Menu. All installed Channels have identical screen features and functions.

To setup an Occlusion Test, Press the **Occlusion Test** key from the selected Channel's Main Menu. The following Occlusion Test setup screen will appear:

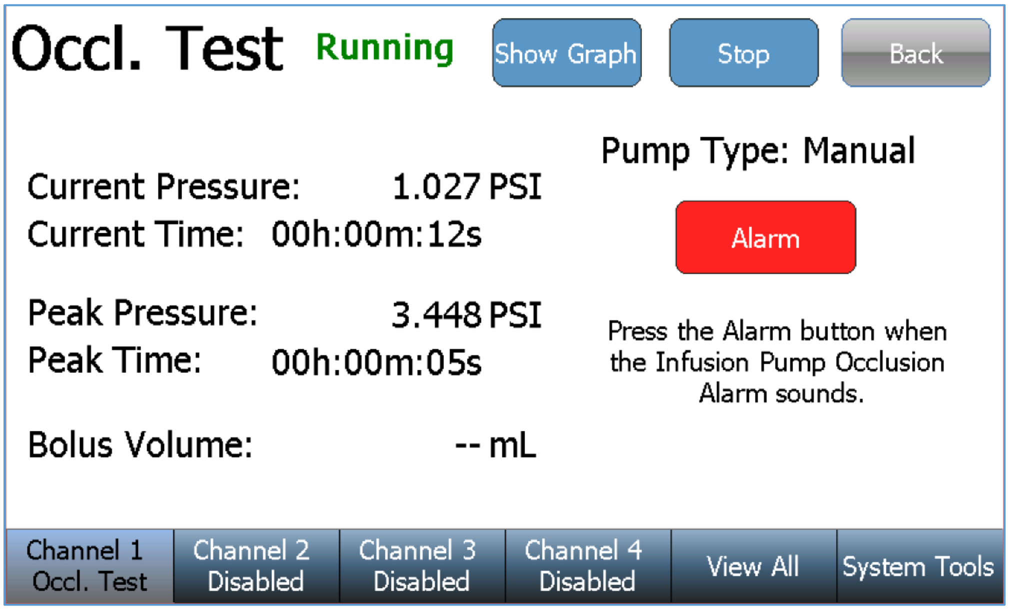


The Occlusion Test setup screen includes the occlusion test setup and control parameters. To change a setting, simply press the parameter to display the available options, select an option, or enter the desired value. Use the **Save Setup** key to save the current selections for use with another Channel or another IPA-3400. See the “Loading a Test Template” section of this manual for details.

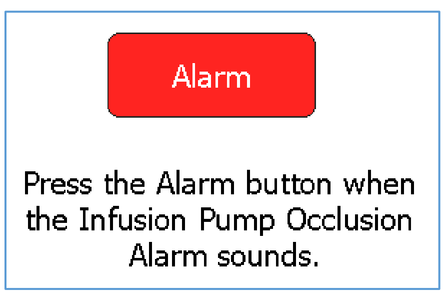
Parameter	Range	Units
Pump Type	Manual	------
Start Condition*	Automatic, Manual	------
Test Limits:	------	------
(Test Parameter)	Pressure, None	------
(Upper Limit)	0 to 50	PSI
(Lower Limit)	0 to 50	PSI

* Automatic start indicates that a test will begin when the **Start** key is pressed and flow is detected. Manual start indicates that the test will start immediately when the **Start** key is pressed.

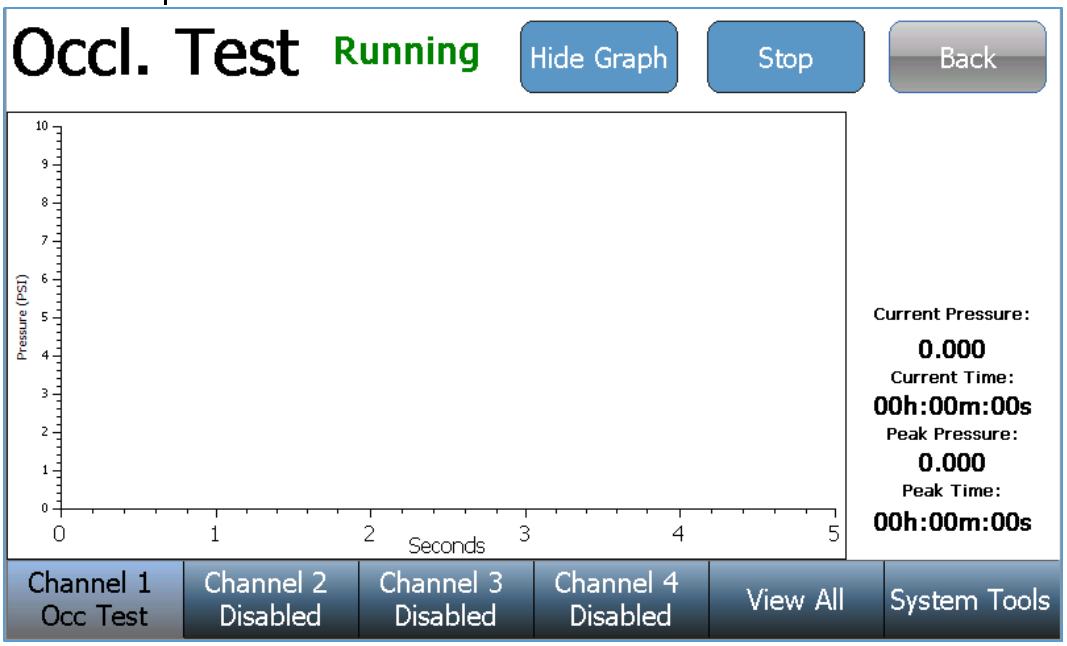
Once the desired conditions have been set, press the **Start** key to initiate the Occlusion Test. The screen will change to the following:



Begin an infusion with the connected pump. Once the IPA-3400 senses pressure from the infusion pump, the IPA-3400 **Alarm** key will enable (change to a red color, as shown below). To complete the occlusion test, press the IPA-3400 **Alarm** key immediately after the infusion pump occlusion alarm occurs. The IPA-3400 will then release the occlusion and drain any infused fluid still in the line.

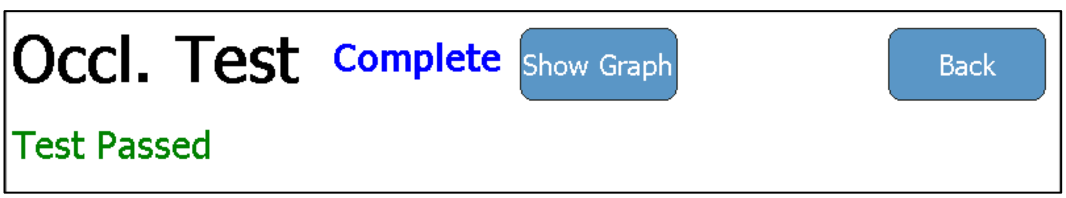


The **Show Graph** key may be used at any time during or after the test to change the Screen to the Graph mode as shown below:

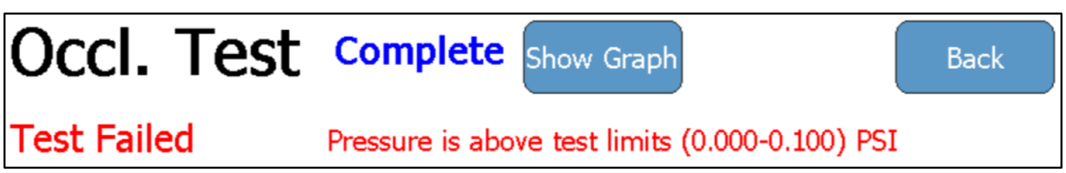


Use the **Hide Graph** key to return to the previous display.

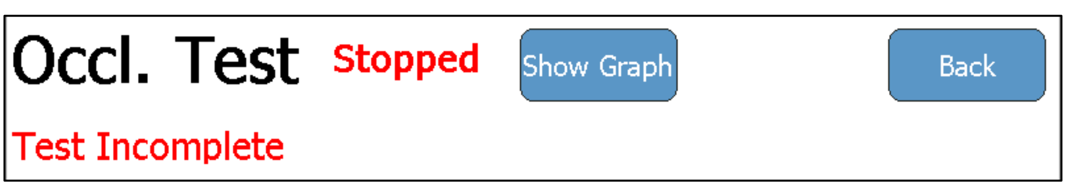
Upon test completion, the test status will change from **Running** to **Complete** and the pass/fail test result will be displayed:



If the test has failed, a brief description of the failure is included:



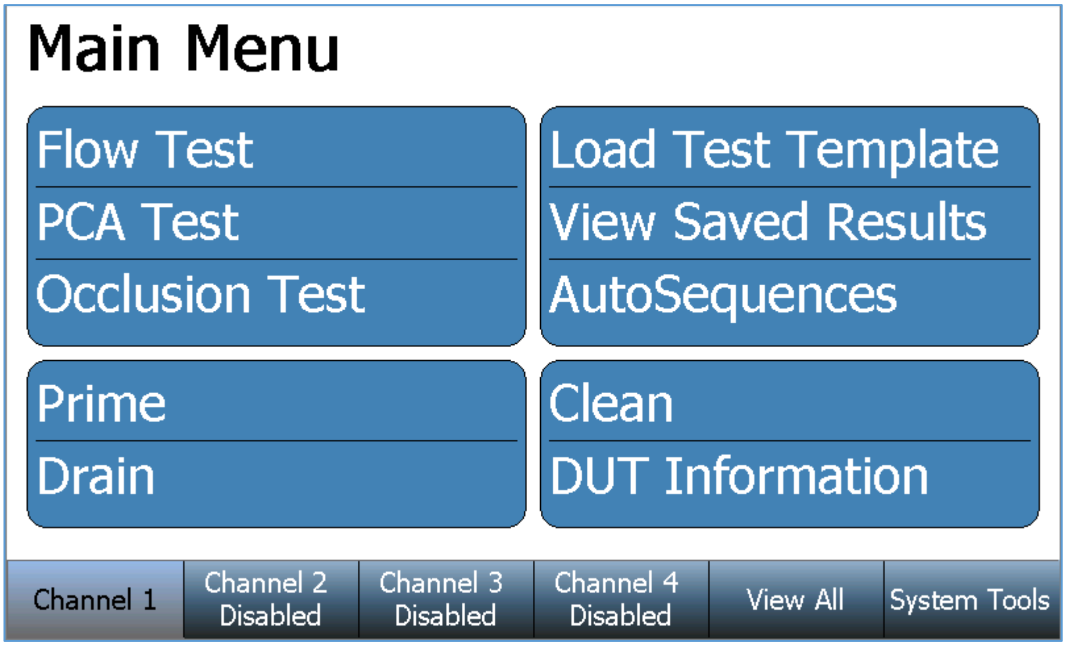
If the test is manually stopped, the test result will change from **Running** to **Stopped** and the test result will indicate an incomplete test:



AUTOSEQUENCES

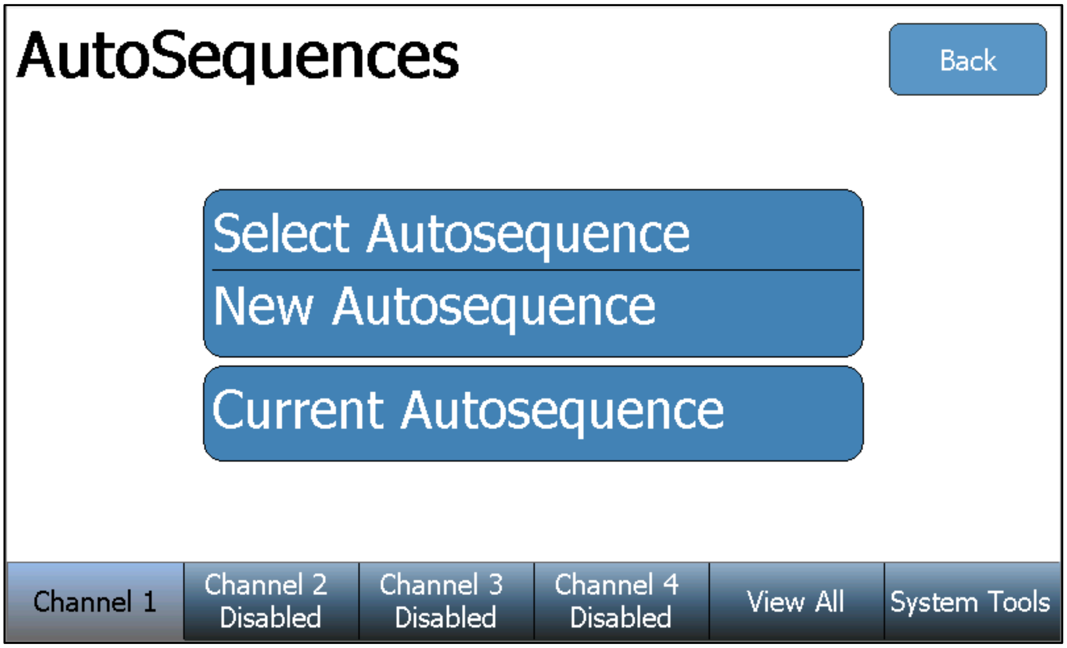
Creating an AutoSequence

An AutoSequence is a repeatable set of consecutive tests to be performed on a single device. The AutoSequence can consist of any number or combination of Instruction Steps, Flow Tests, PCA Tests, and Occlusion Tests. The AutoSequence may be created to incorporate all necessary steps of a device OEM's recommended maintenance procedure. The AutoSequence also prompts the user to enter the device information, providing a complete maintenance record.

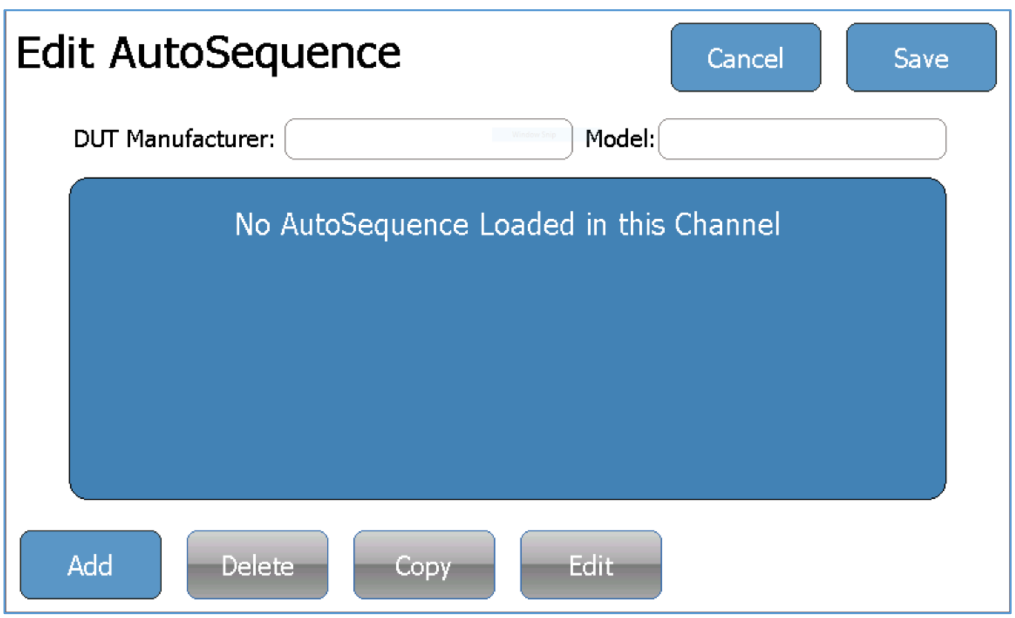


Above is the Main Menu screen for a selected Channel. If needed, press the **Back** key on any successive screen to return to the Channel's Main Menu. All installed Channels have identical screen features and functions.

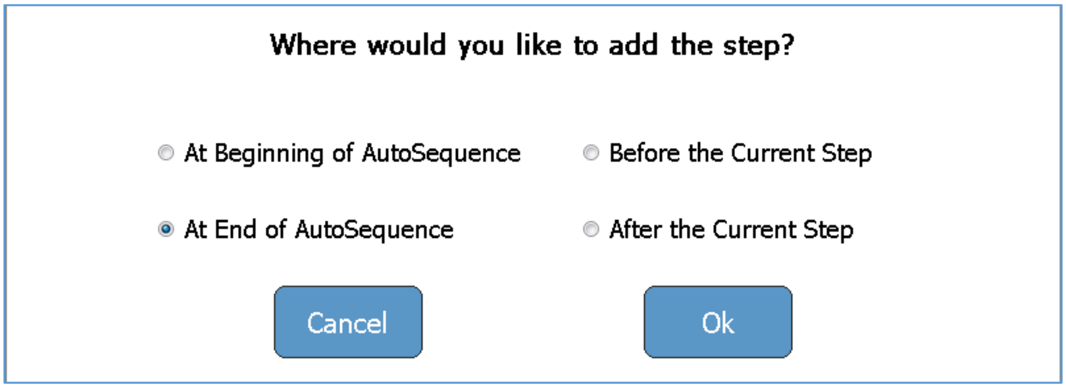
To create an AutoSequence Press the **AutoSequences** key and the following screen will appear:



Press the **New AutoSequence** key to open a blank AutoSequence summary (as shown below). Press the **Add** key from this summary screen to create a step of the AutoSequence.

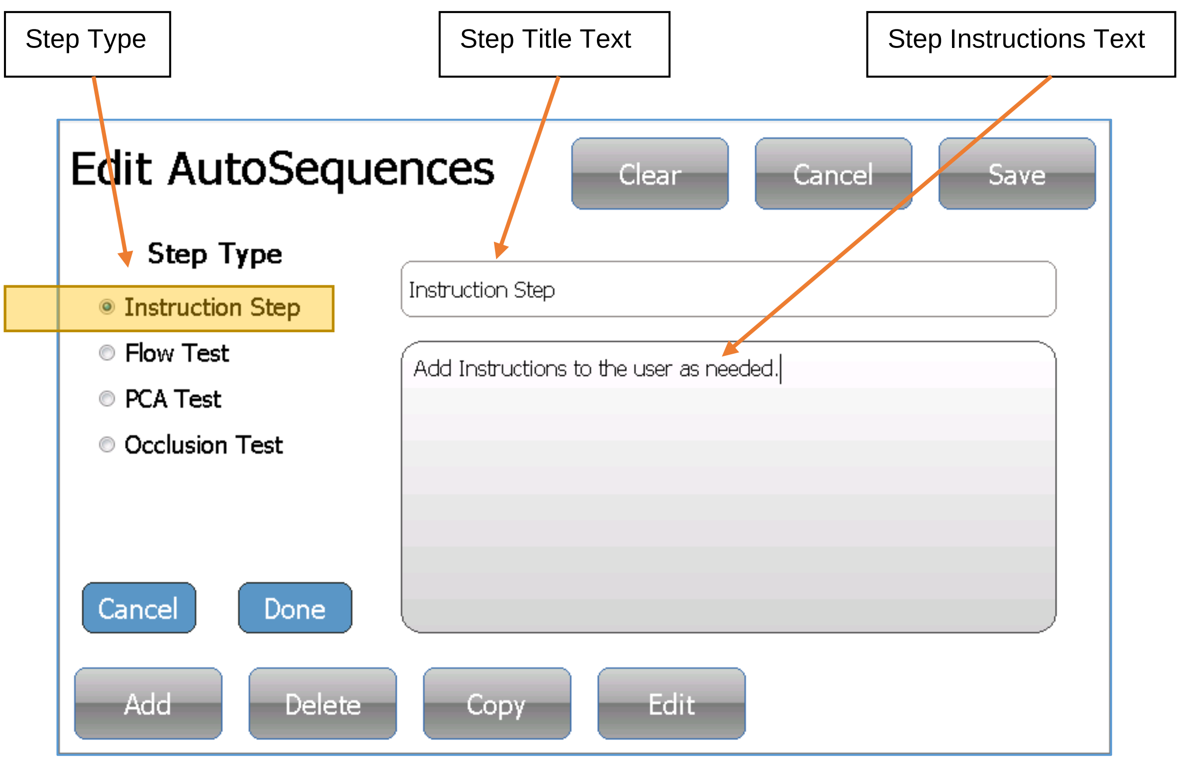


If the AutoSequence already contains a step, the IPA-3400 will prompt the user to select the desired location of the new step.



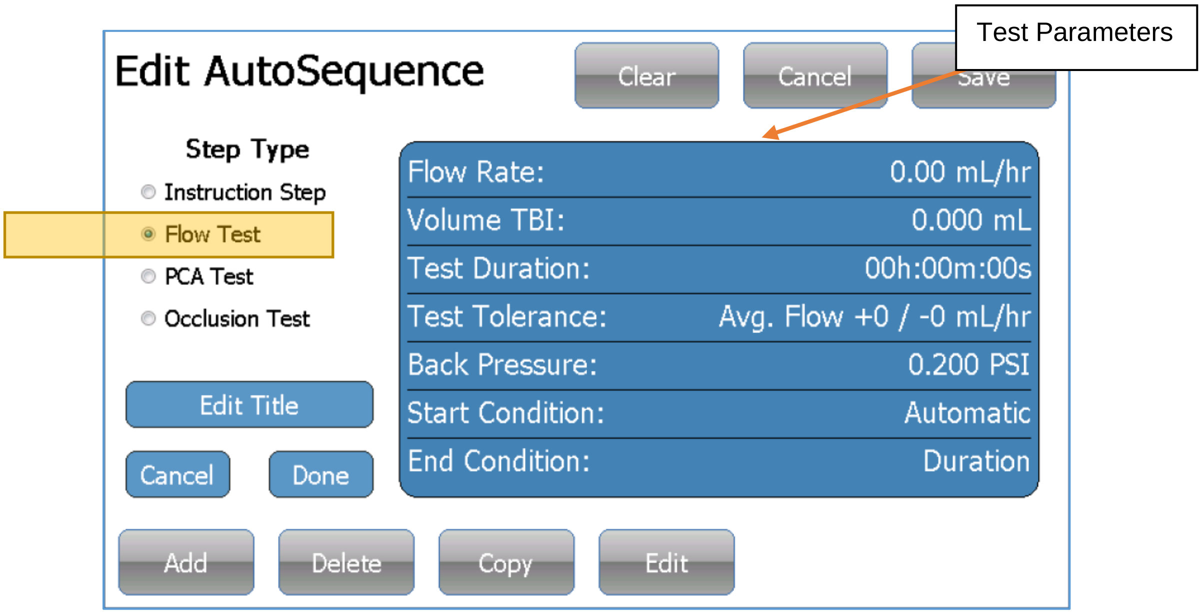
Begin the step setup by selecting the Step Type (Instruction Step, Flow Test, PCA Test, or Occlusion Test). Examples of each step type are:

Instruction Step
The Step Title and Step Instructions may be updated using an attached USB keyboard. Simply select a text field on the screen and input the new information from the keyboard. Many OEM maintenance procedures require several physical inspections of the infusion pump components. Instruction Steps are a convenient way to include these inspections in the AutoSequence, as part of the complete maintenance record. After the Instruction Step information has been updated, select the **Done** key to return to the AutoSequence summary screen.



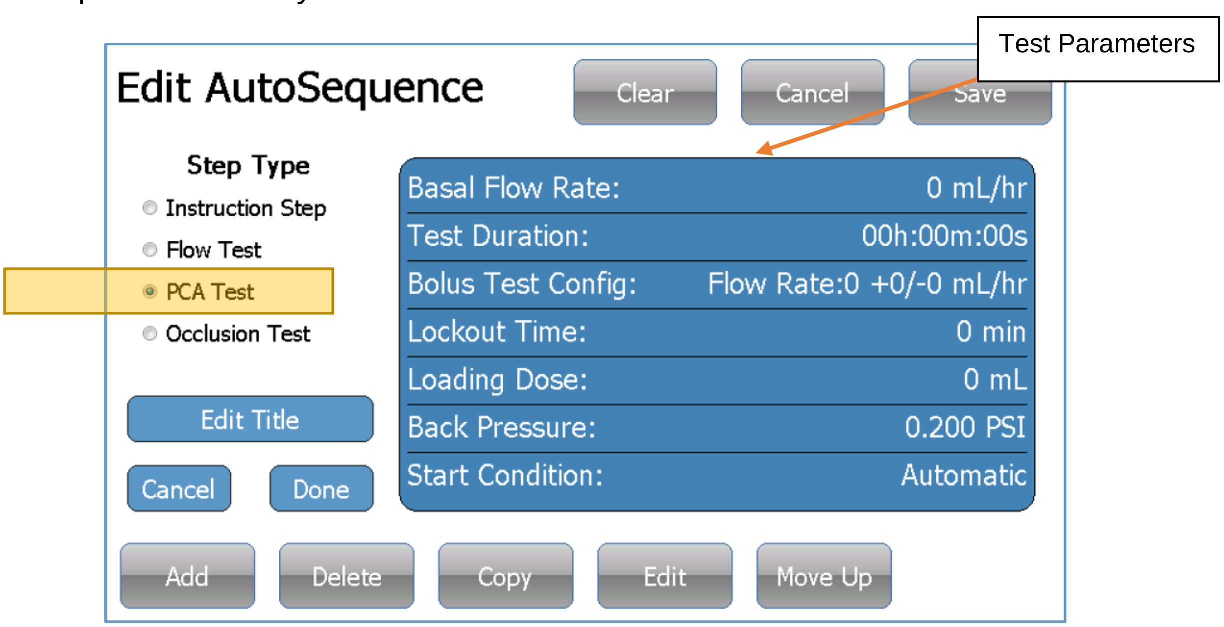
Flow Test

The Flow Test step parameters are identical to an individual Flow Test. See the "Running a Flow Test" section of this user manual for details. The Edit Title button can be used to change the Title displayed in the summary of AutoSequence steps if desired. After the Flow Test Step parameters have been updated, select the **Done** key to return to the AutoSequence summary screen.



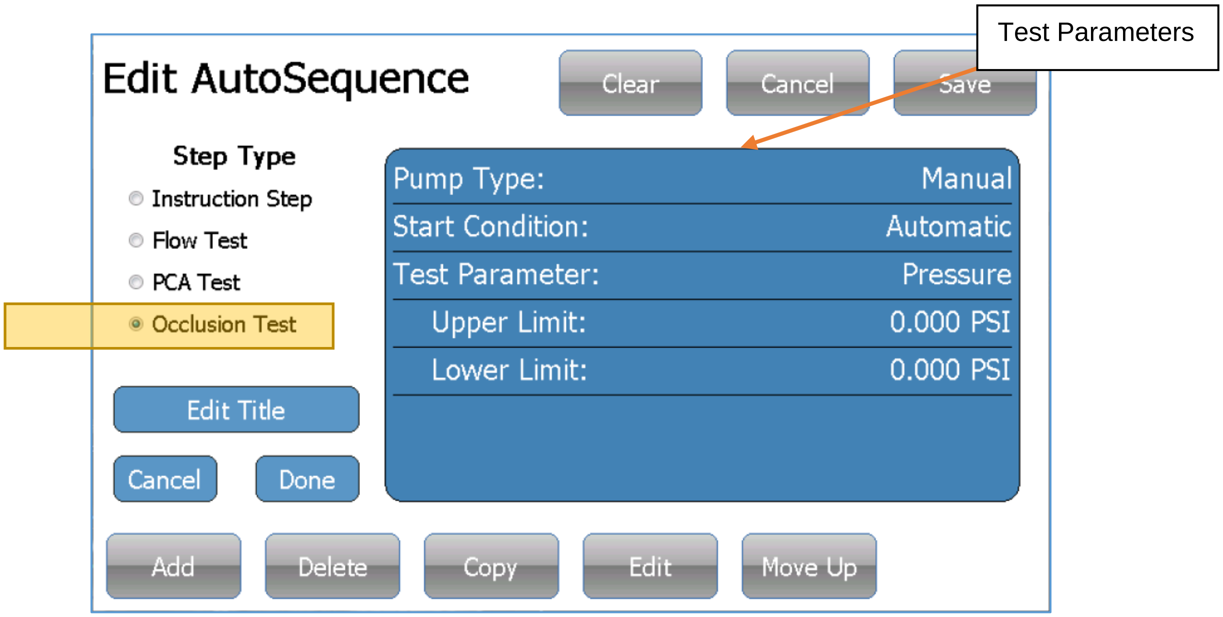
PCA Test

The PCA Test step parameters are identical to an individual PCA Test. See the "Running a PCA Test" section of this user manual for details. The Edit Title button can be used to change the Title displayed in the summary of AutoSequence steps if desired. After the PCA Test Step parameters have been updated, select the **Done** key to return to the AutoSequence summary screen.



Occlusion Test

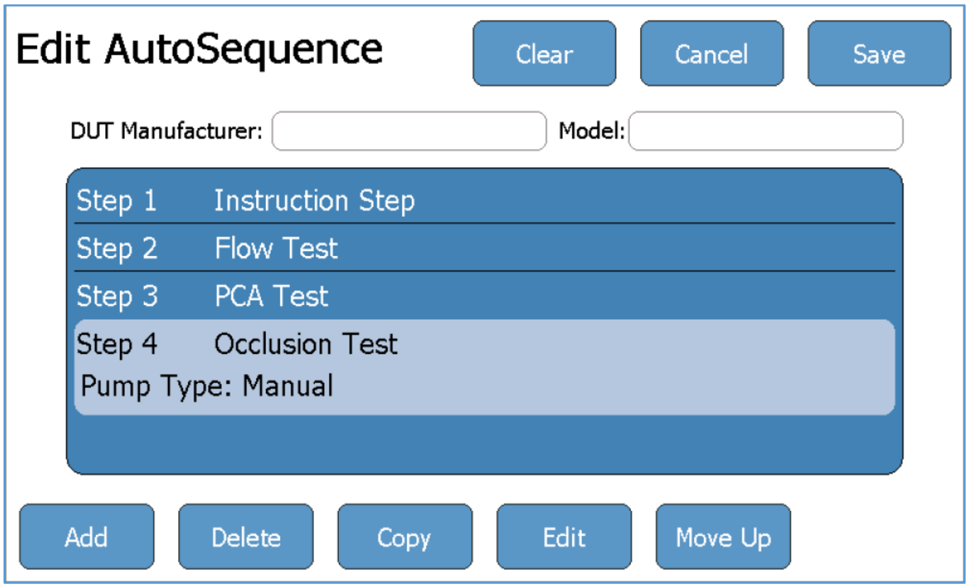
The Occlusion Test step parameters are identical to an individual Occlusion Test. See the "Running an Occlusion Test" section of this user manual for details. The Edit Title button can be used to change the Title displayed in the summary of AutoSequence steps if desired. After the Occlusion Test Step parameters have been updated, select the **Done** key to return to the AutoSequence summary screen.



From the AutoSequence summary screen, the user may also use the **Delete**, **Copy**, **Edit**, **Move Down**, and **Move Up** keys. These keys will respectively remove, copy, edit, or rearrange the highlighted AutoSequence step.

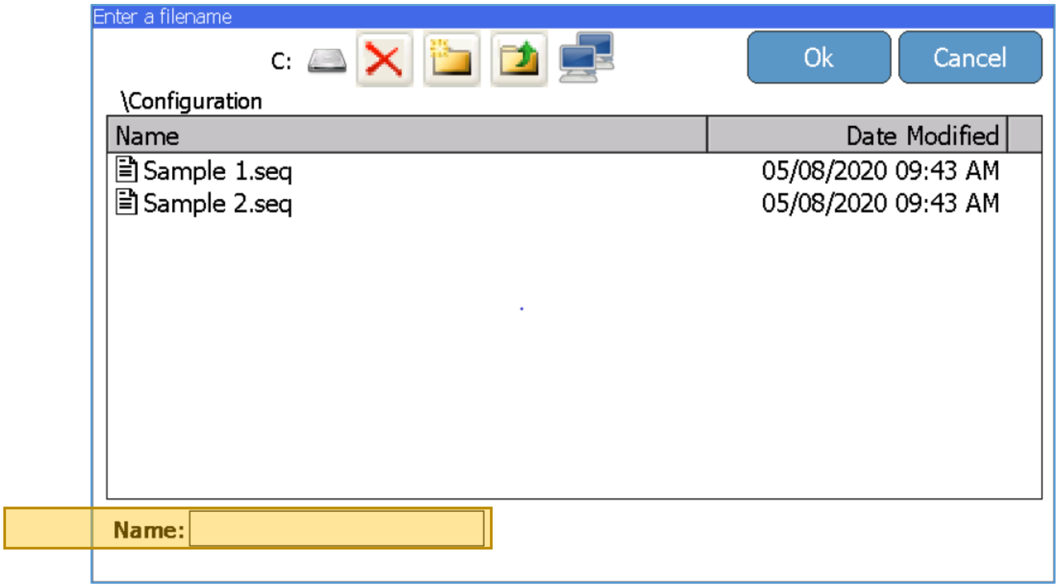
Some AutoSequences are designed for specific Manufacturer and Models of Infusion Pumps. The optional DUT Manufacturer and Model options allow this information to be stored in the AutoSequence and this information will be automatically filled into the DUT Information panel when the AutoSequence is run.

Once the AutoSequence summary screen contains all required steps in the desired order, press the **Save** key.



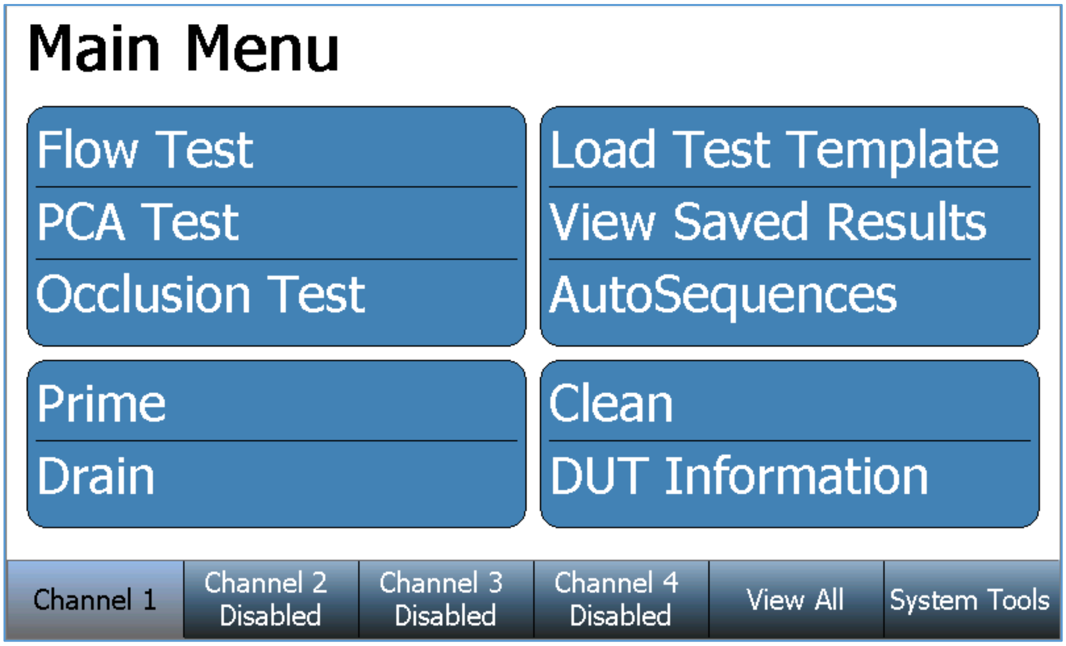
Enter the new AutoSequence filename and press the **Enter** key.

Verify the correct filename and file storage location, then press the **Ok** key to save the AutoSequence.



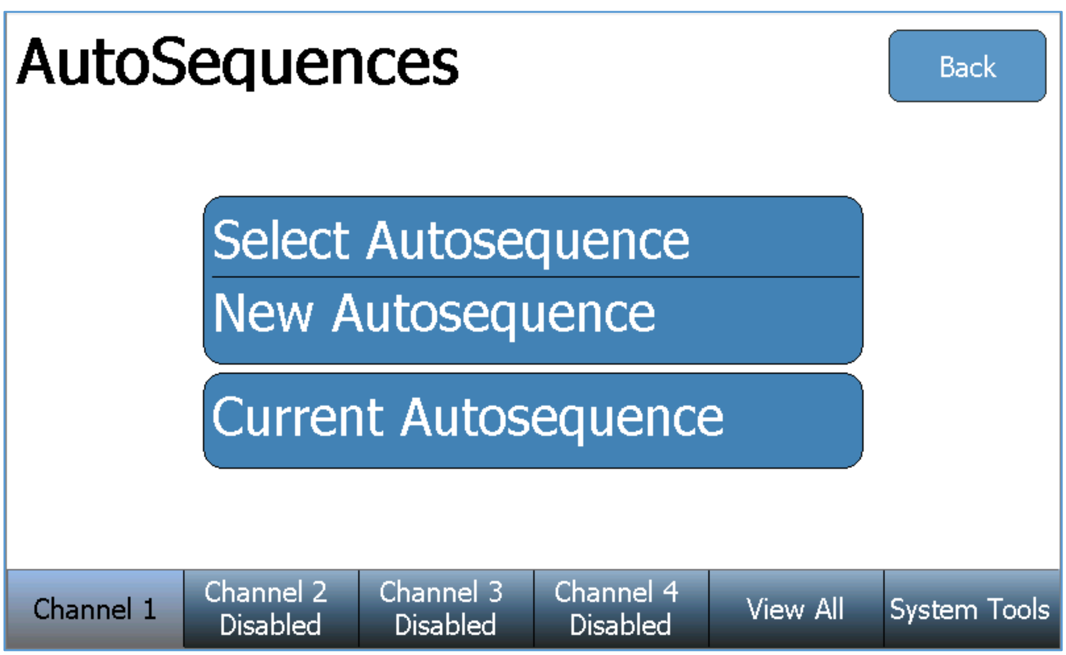
After pressing **Ok**, the IPA-3400 will return to AutoSequence View screen.

Running an AutoSequence

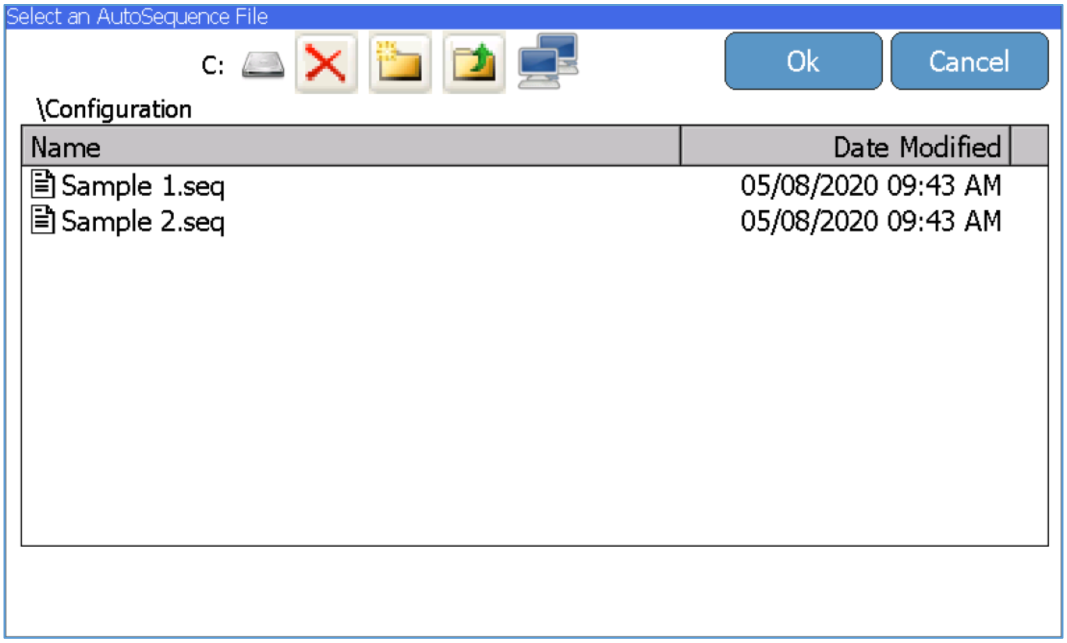


Above is the Main Menu screen for a selected Channel. If needed, press the **Back** key on any successive screen to return to the Channel's Main Menu. All installed Channels have identical screen features and functions.

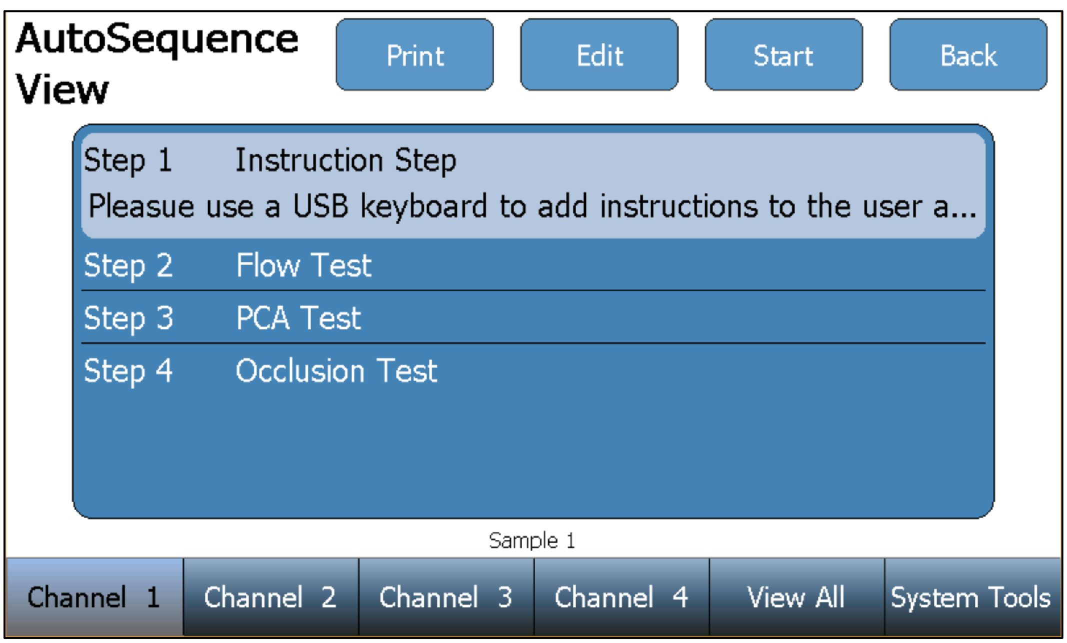
To run an AutoSequence Press the **AutoSequences** key and the following screen will appear:



Press the **Select AutoSequence** key to use a previously created/saved AutoSequence file. The file browser window will appear and allow access to all saved AutoSequence files on the IPA-3400 (C: drive), or any inserted USB flash drive (D:, E:, F:, G: drives).

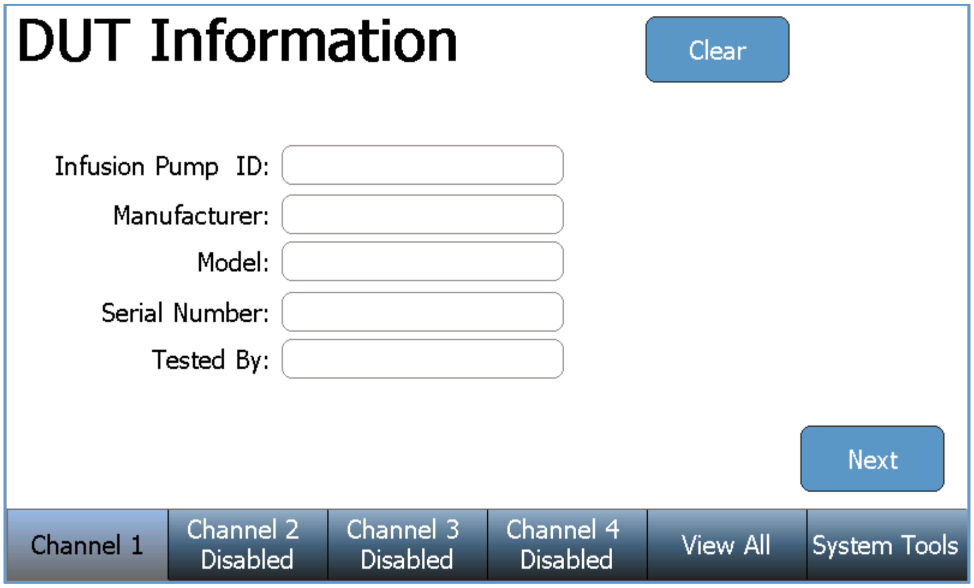


Once an AutoSequence file has been selected, the IPA-3400 will load all test steps and parameters. The AutoSequence summary will be displayed, similarly to the screen below:



Each individual AutoSequence step may be selected and reviewed/edited using the **Edit** key. See the “Creating an AutoSequence” section of this manual for editing details.

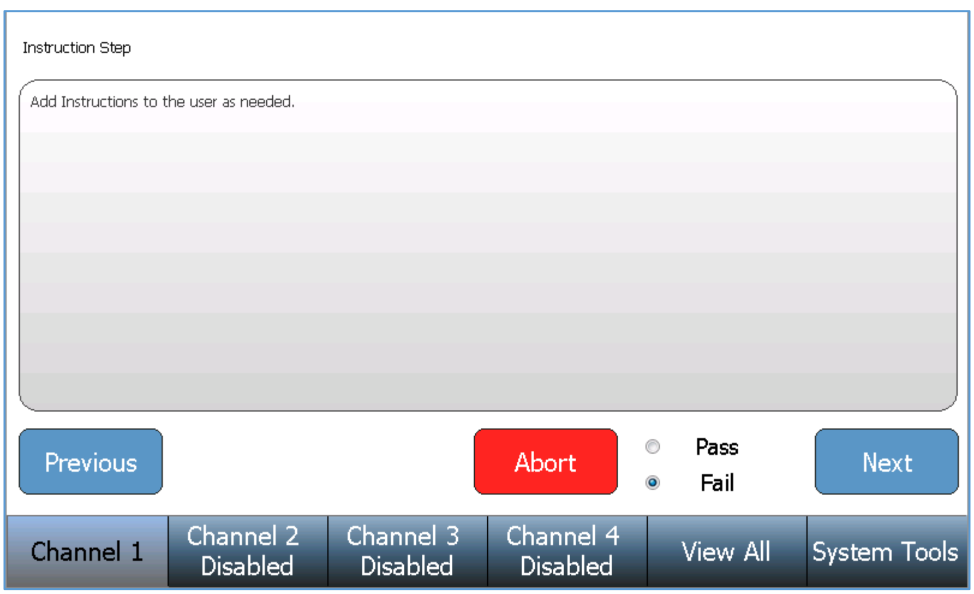
Press the **Start** key to perform the AutoSequence testing. The IPA-3400 will prompt the user for the infusion pump (DUT) information:



After the information has been entered, press the Next key to begin Step 1 of the AutoSequence. The AutoSequence may contain any number of four different test types (Instruction Step, Flow Test, PCA Test, and Occlusion Test). Each test type will require interaction with the user before the IPA-3400 will proceed to a subsequent step. Examples of each test type and the required user interaction are below:

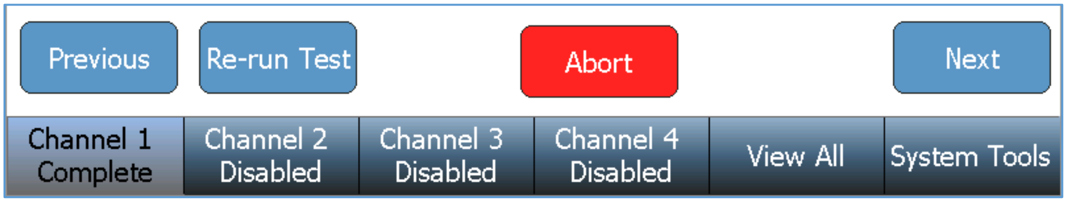
Instruction Step

The user must follow the written instructions, select the Pass or Fail condition, and press next to proceed to the next AutoSequence step.

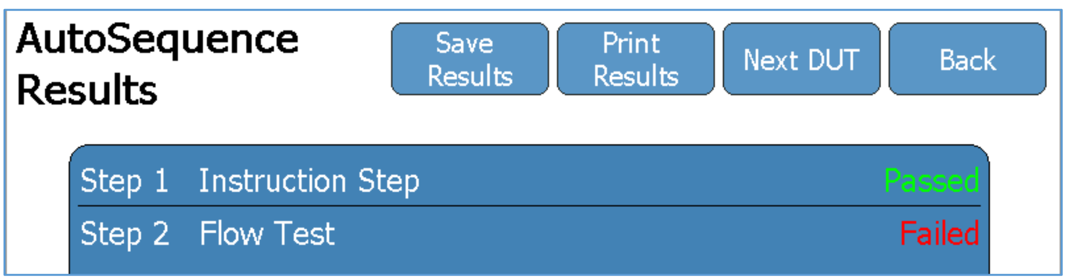


Flow Test, PCA Test, and Occlusion Test

Each of these steps are performed and completed the same as an individual Flow, PCA, or Occlusion test. See the "Running a Flow Test", "Running a PCA Test", and "Running an Occlusion Test" sections for test details. Once a step has completed a **Re-run Test** key, an **Abort** key, and a **Next** key will appear near the bottom of the screen. The test may be repeated using the **Re-run Test**. The AutoSequence may be aborted (losing all test data) using the **Abort** key. Proceed to the next AutoSequence step using the **Next** key.

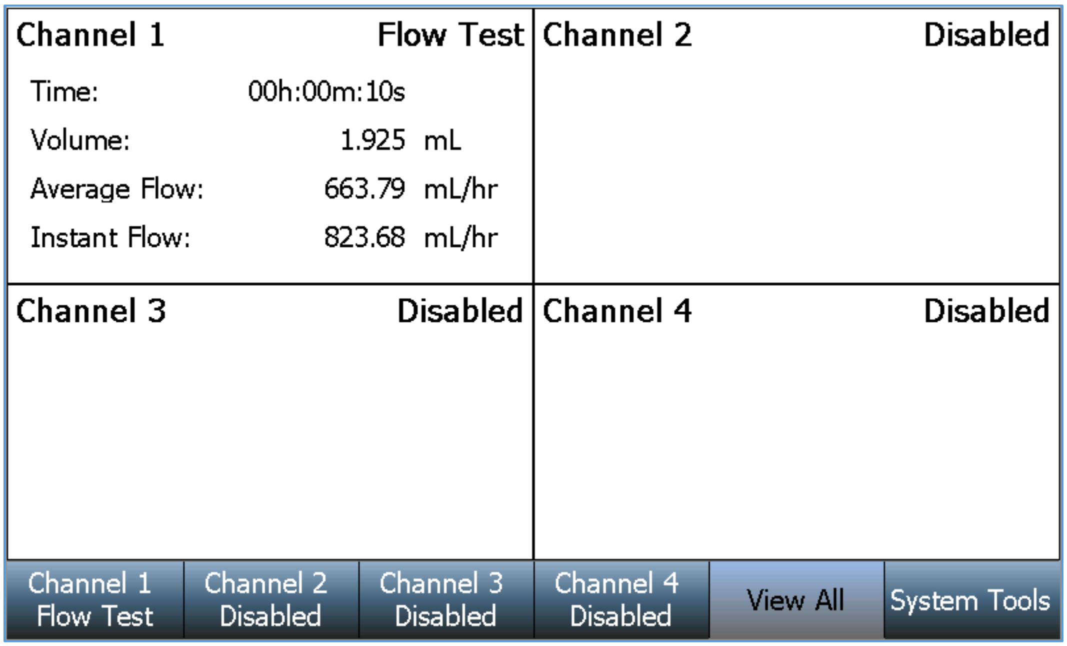


After completion of the final step, an AutoSequence Results summary is provided with a pass/fail status of each step. The AutoSequence Results may be saved and/or printed using the **Save Results** key and **Print Results** key near the top of the screen. Pressing the **Next DUT** key allows the user to perform the same AutoSequence on another infusion pump, beginning the AutoSequence at the DUT Information screen.



View All

A View All screen is provided that allows the user to simultaneously view the data of all four Channels. The data displayed for each Channel is a subset of the information that appears in the individual Channel screens. Pressing the **View All** key changes the screen to the View All display:



All four Channels will be displayed, regardless of the number of Flow Modules installed. If there is no Flow Module installed for a Channel, the display will indicate the Channel as Disabled.

If not in use, a Channel will be shown as Idle.

If a Test is being run on a Channel, that Channel will display Test information.

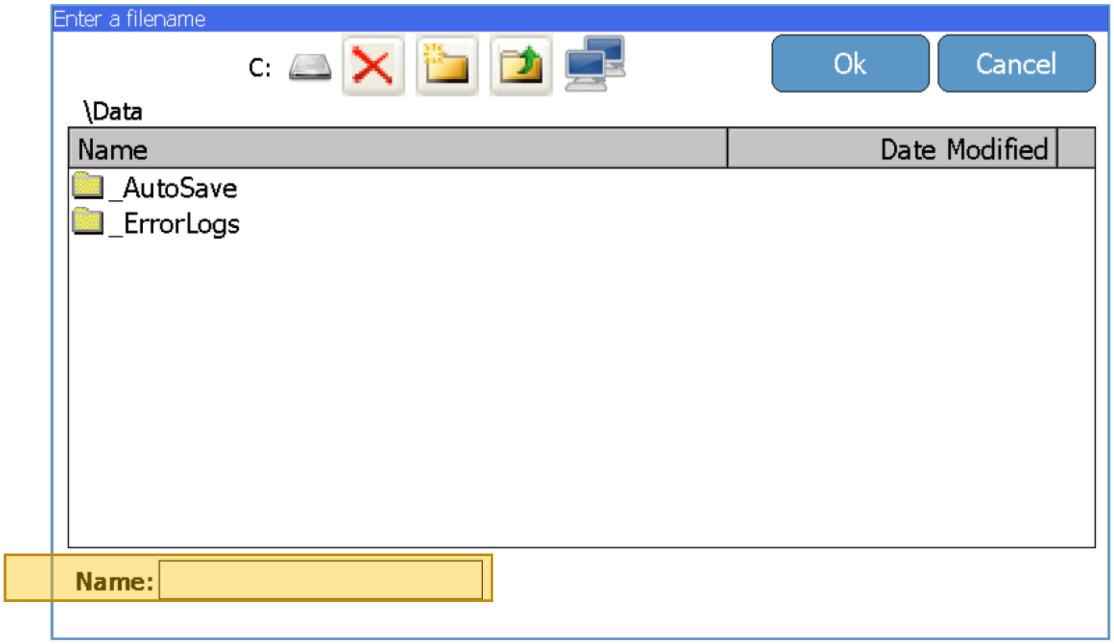
Saving and Viewing Test Results

Saving Test Results

Test results may be saved by pressing the **Save Results** key that appears on the screen at the completion of a test.

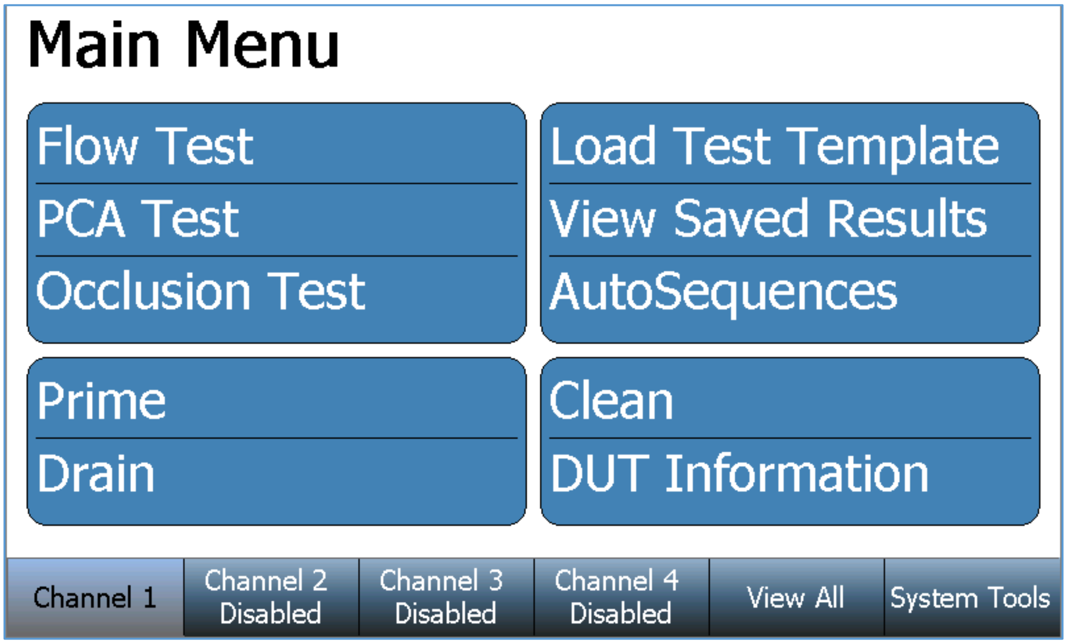
Enter the test results filename and press the **Enter** key.

Verify the correct filename and file storage location, then press the **Ok** key to save the test results.



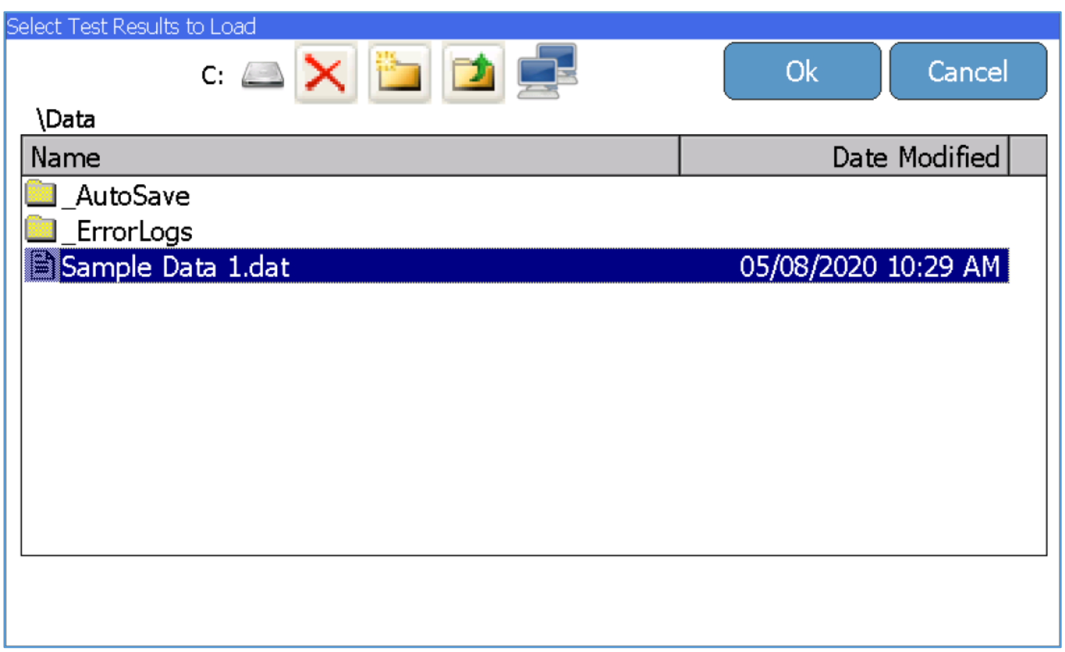
After pressing **Ok**, the IPA-3400 will return to its previous screen.

Viewing Test Results

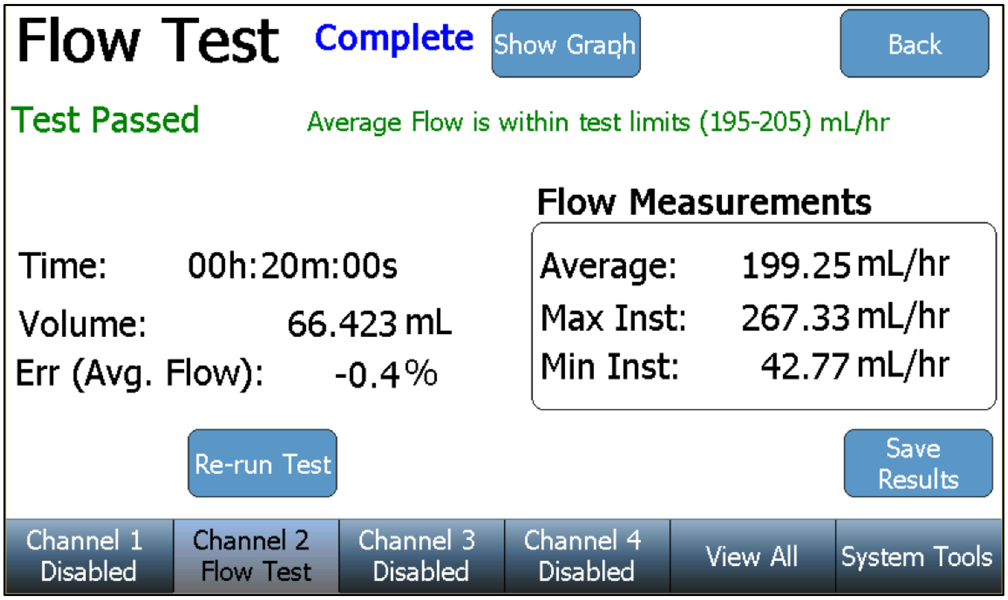


Above is the Main Menu screen for a selected Channel. If needed, press the **Back** key on any successive screen to return to the Channel's Main Menu. All installed Channels have identical screen features and functions.

Press the **View Saved Results** key from Main Menu screen to display the saved results files. The file browser window will appear and allow access to all saved test results files on the IPA-3400 (C: drive), or any inserted USB flash drive (D:, E:, F:, G: drives).



Select the desired file and press the **Ok** key. The display will repopulate all test data and return to the test results screen, allowing the user to view the data and graph.



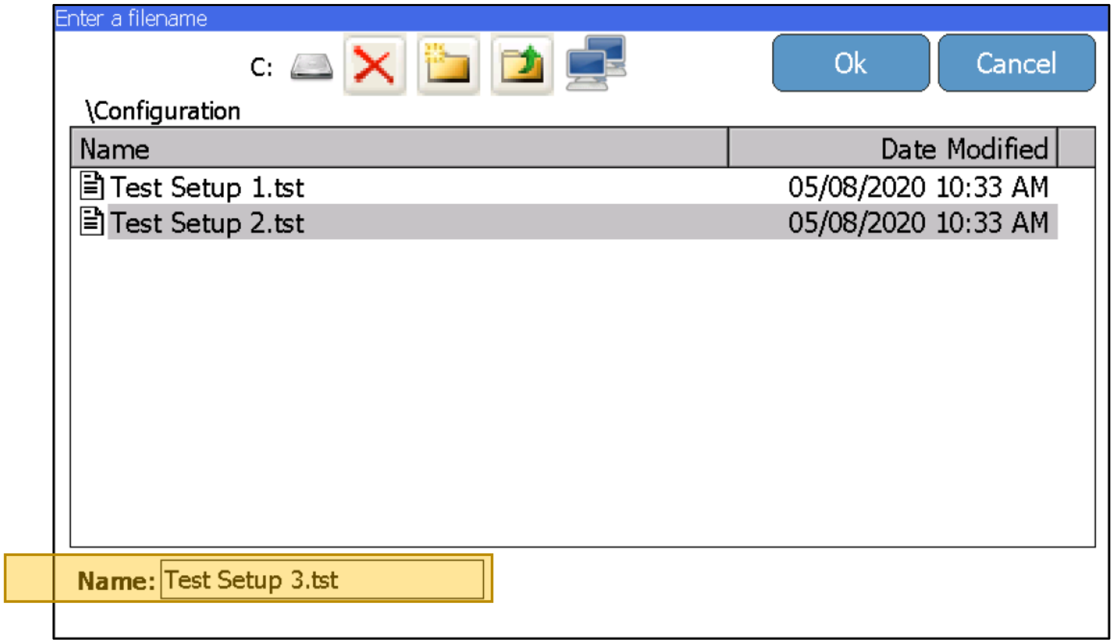
Saving and Loading Test Setups

Saving Test Templates

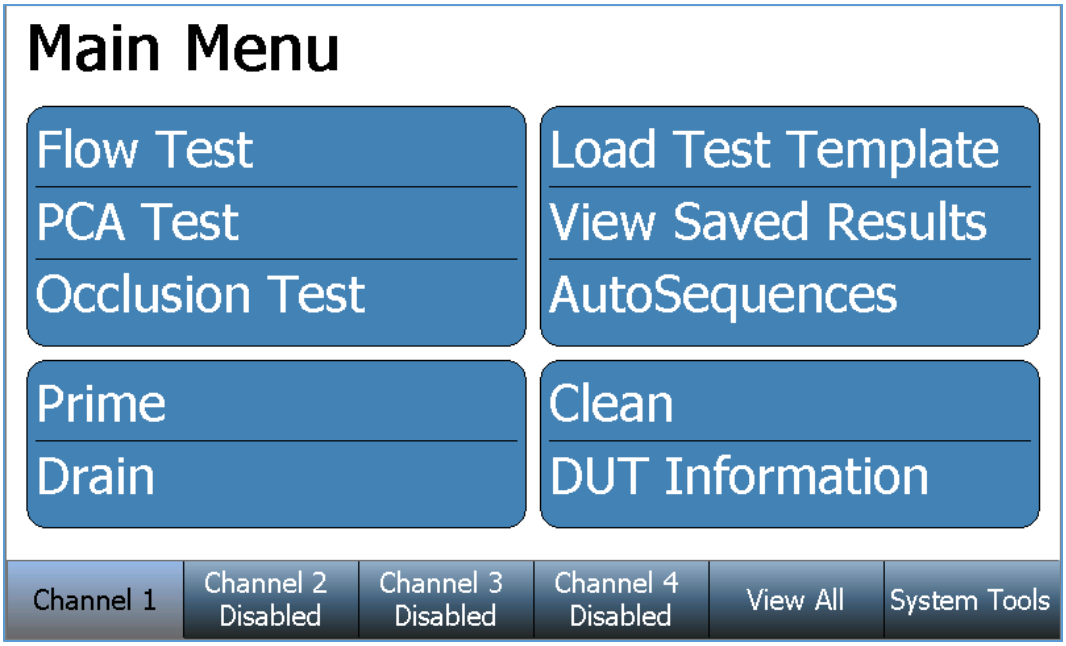
Individual test setups may be saved for later use or as a template for similar individual tests. From the **Flow Test, PCA Test,** or **Occlusion Test** setup screen press the **Save Setup** key.

Enter the new test setup filename and press the **Enter** key.

Verify the correct filename and file storage location, then press the **Ok** key to save the test setup.

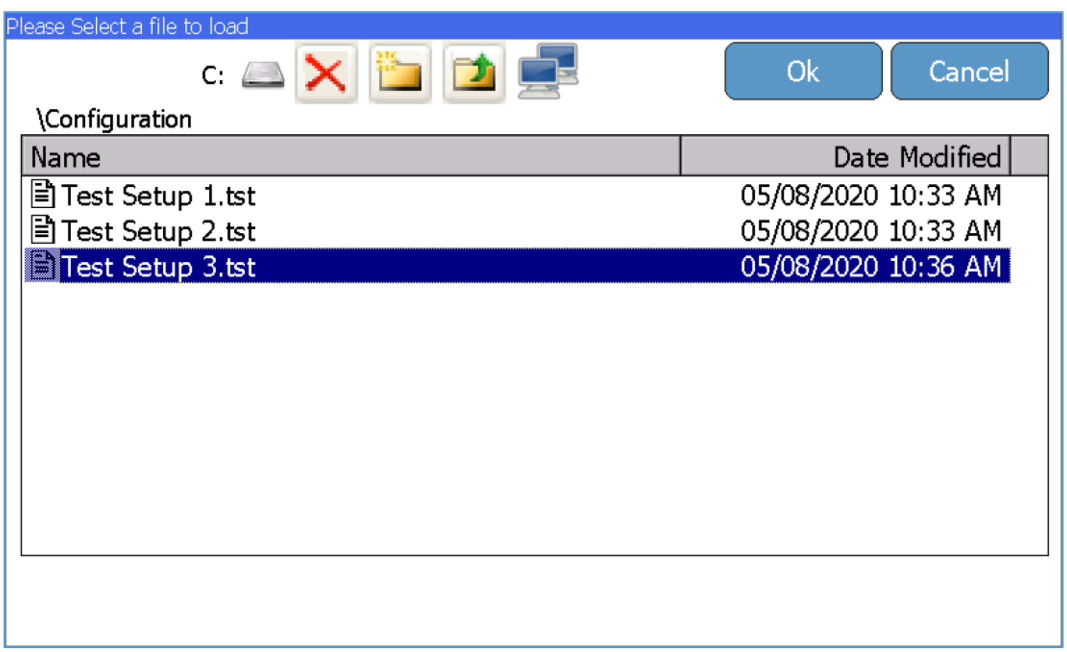


Loading a Test Template

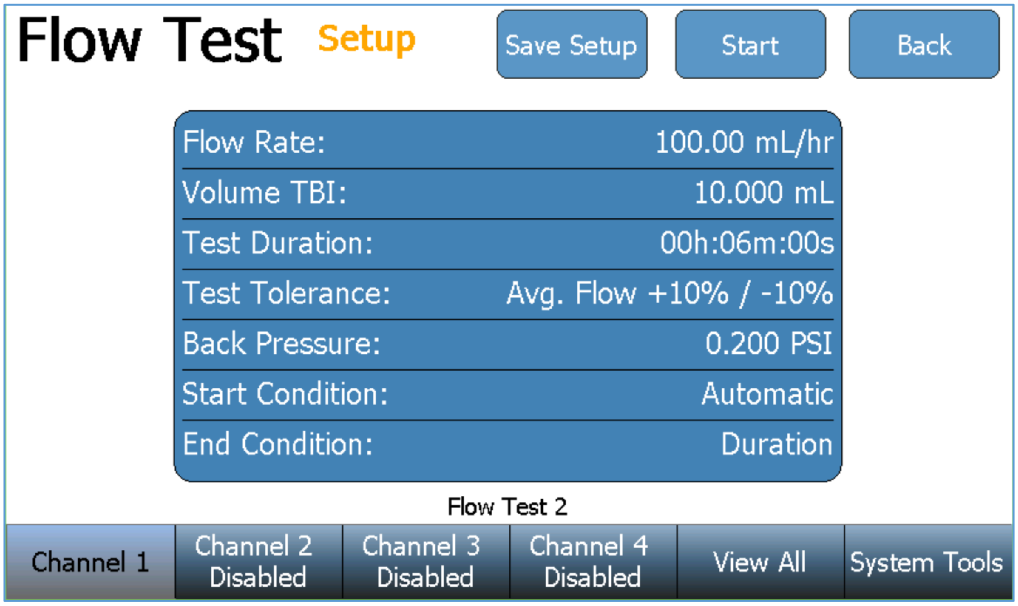


Above is the Main Menu screen for a selected Channel. If needed, press the **Back** key on any successive screen to return to the Channel's Main Menu. All installed Channels have identical screen features and functions.

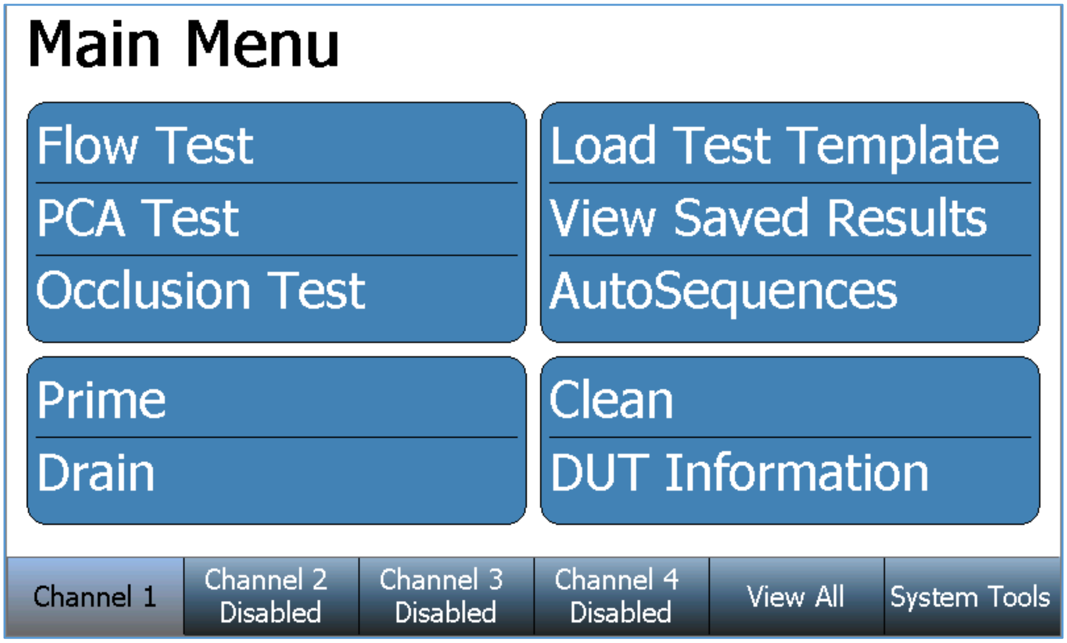
Press the **Load Test Template** key to use a previously created/saved test setup. The file browser window will appear and allow access to all saved test template files on the IPA-3400 (C: drive), or any inserted USB flash drive (D:, E:, F:, G: drives).



Select the desired file and press the **Ok** key. The display will repopulate all test parameters and return to the test setup screen, allowing the user to edit the parameters and/or start the test.



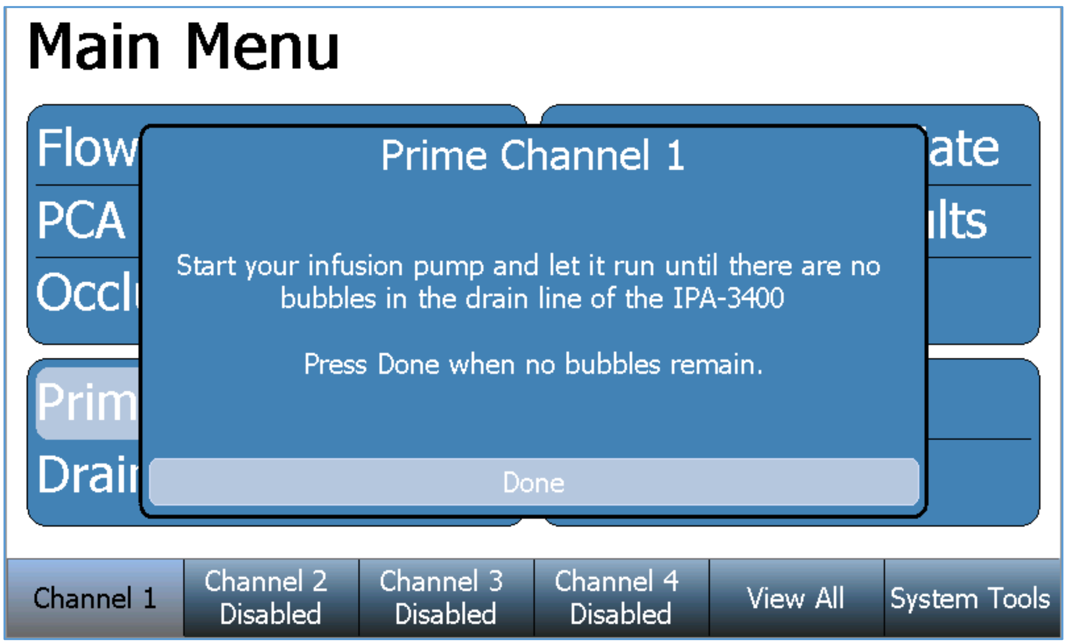
Prime, Drain, and Clean Functions



Above is the Main Menu screen for a selected Channel. If needed, press the **Back** key on any successive screen to return to the Channel's Main Menu. All installed Channels have identical screen features and functions.

Prime

From the selected Channel's Main Menu press the **Prime** key. The following instruction window will appear and the Flow Module will begin priming the connected tubing. Start the infusion pump to allow fluid to dispense through the Flow Module. Note: A higher infusion pump flow rate will expedite the priming process.



Press the **Done** key when no visible bubbles remain in the drain line tubing.

Drain

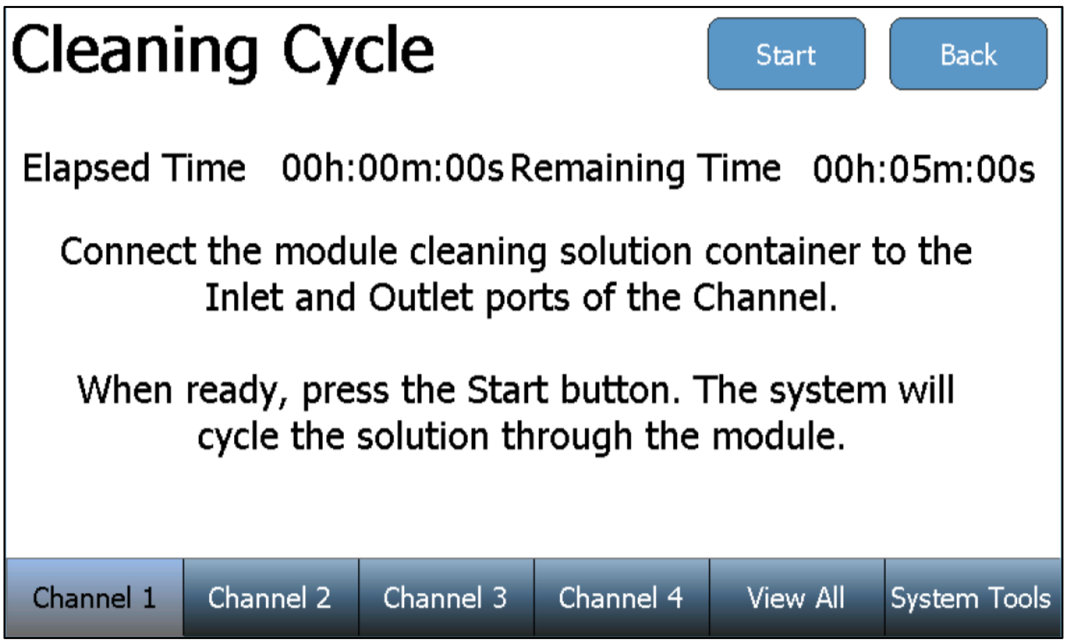
From the selected Channel's Main Menu press the **Drain** key. The following instruction window will appear and the Flow Module will begin draining.

Press the **Done** key when no fluid remains in the drain line tubing.

Clean

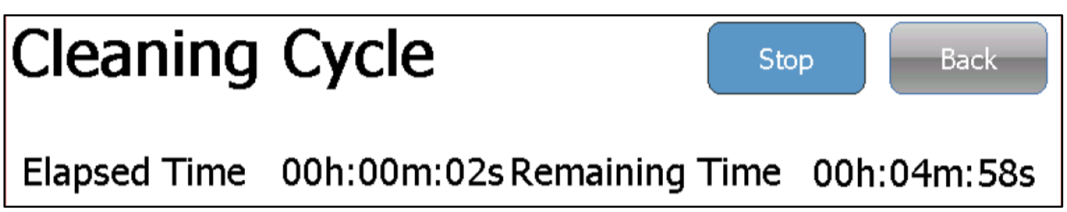
Flow Module cleaning must be performed periodically to remove any possible contaminants or debris in the flow and measurement path. The recommended cleaning solution is “MICRO-90 ®”, available from Cole-Parmer ® (www.coleparmer.com). To prepare the 1% test solution, mix 10 mL of “MICRO-90 ®” per 1 Liter of Distilled Water. If foaming becomes a problem, change the ratio to 5 mL of “MICRO-90 ®” per 1 Liter of Distilled Water.

From the selected Channel’s Main Menu press the **Clean** key. The following screen will appear:



Follow the displayed instructions, then press the **Start** key.

The Flow Module will perform its cleaning cycle until the Remaining Time expires, or the user presses the **Stop** key.

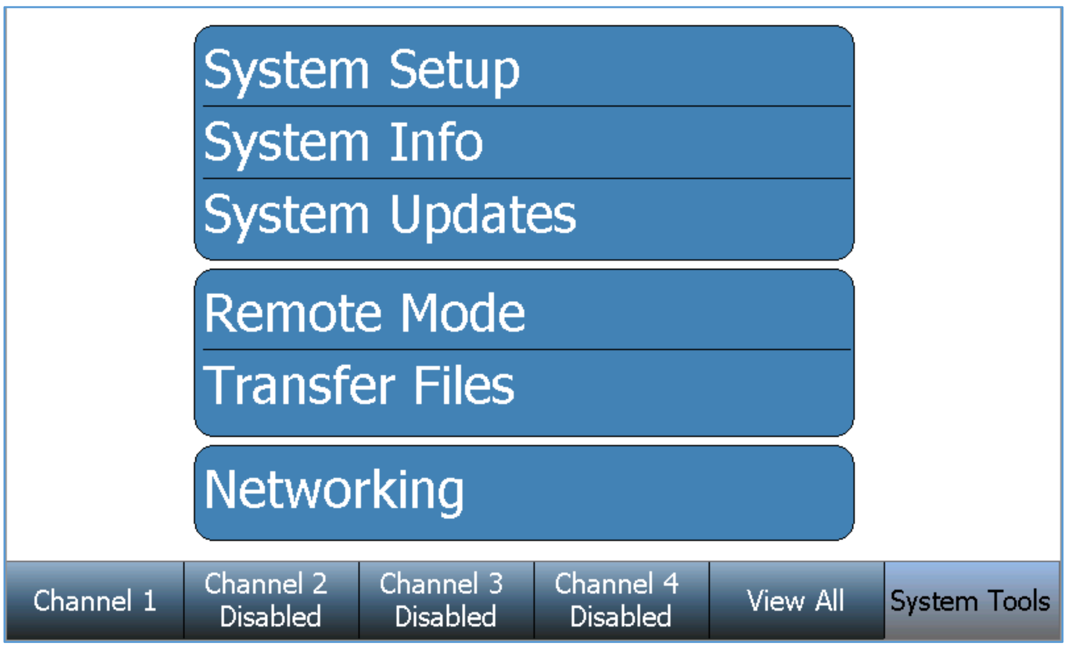


System Tools

The System Tools screen contains the IPA-3400 general setup, information, and special functions for the system.

Press the **System Tools** key at the bottom of the screen:

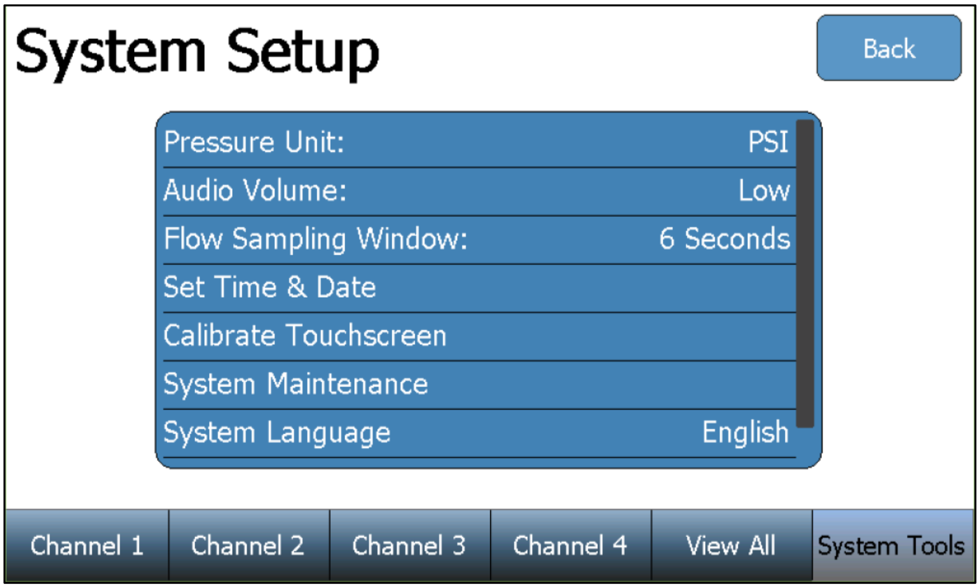
The following System Tools screen will appear:



Press any of the available keys to open the screen for the labeled function.

System Setup

Press the **System Setup** key to display the following screen:

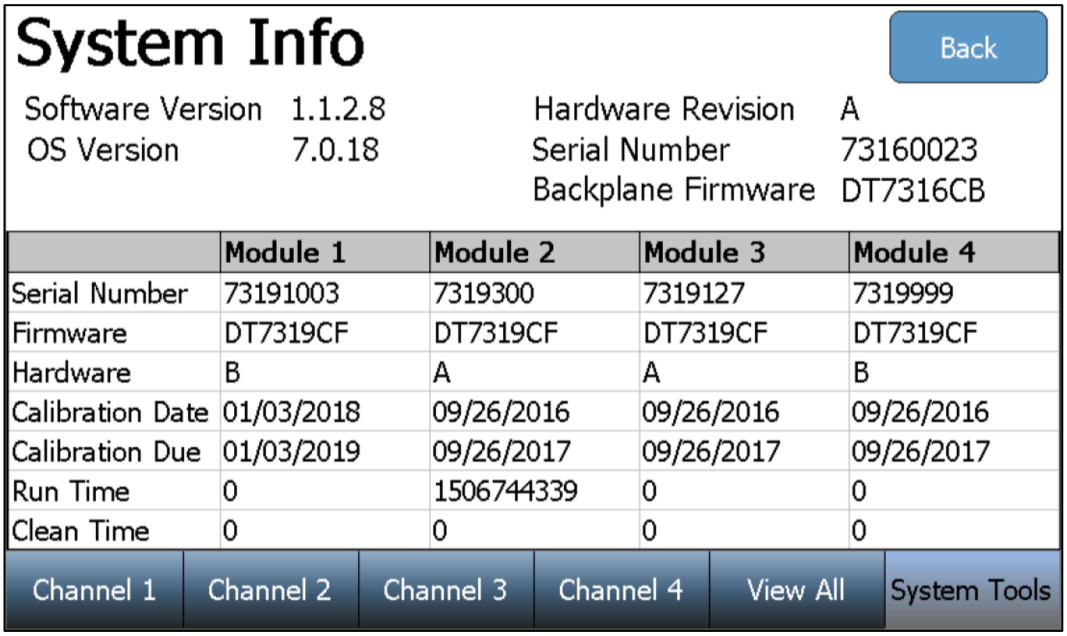


To change a setting, simply press the parameter to display the available options, select an option, or enter the desired value. The scrollbar on the side of the menu indicates that you can drag the menu up or down to browse more options.

Parameter	Choices
Pressure Units	mmHg, PSI, mBar or kPa
Audio Volume	High, Medium, Low, Silent
Flow Sampling Window	Samples averaged for the Instantaneous Flow display. (1-10 Seconds)
Set Time & Date	Time (12/24), Day, Month, Year
Calibrate Touchscreen	This is a function to recalibrate the touchscreen. Follow the instructions on the screen.
System Maintenance	This is a restricted feature
System Language	Sets the Language for the system.
Restore Defaults	Restores all system parameters to the default settings
Show Grams	Enables/Disables Grams display in Flow tests

System Info

Press the **System Info** key to display the following screen:



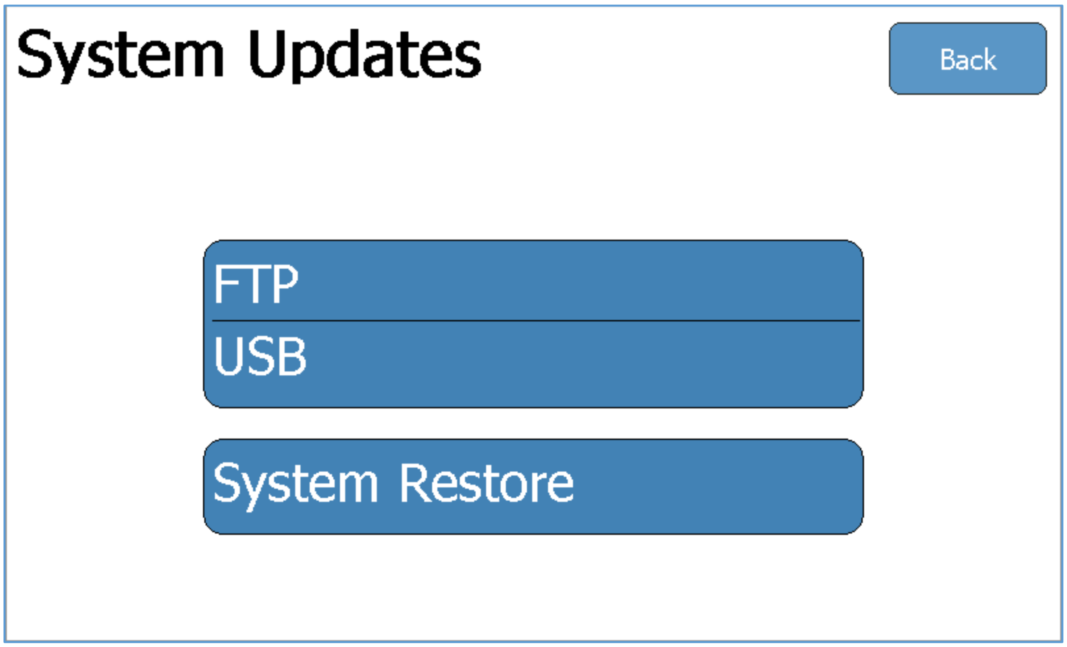
	Module 1	Module 2	Module 3	Module 4
Serial Number	73191003	7319300	7319127	7319999
Firmware	DT7319CF	DT7319CF	DT7319CF	DT7319CF
Hardware	B	A	A	B
Calibration Date	01/03/2018	09/26/2016	09/26/2016	09/26/2016
Calibration Due	01/03/2019	09/26/2017	09/26/2017	09/26/2017
Run Time	0	1506744339	0	0
Clean Time	0	0	0	0

The System Info screen displays the IPA-3400 serial number, software versions, firmware version, and hardware version. The installed Flow Module information is also displayed, including the Serial Number(s), Firmware Version(s), Hardware Version(s), Calibration information, Run Time(s), and Clean Time(s).

Parameter	Choices
Serial Number	Serial number of the unit
Firmware	Firmware currently installed on the Module
Hardware	Product revision of the Module
Calibration Date	Date the Module was last calibrated
Calibration Due	Date the Module is due for calibration
Run Time	An ongoing count of Module usage
Clean Time	A resettable count of Module usage since last time the Clean cycle was run.

System Updates

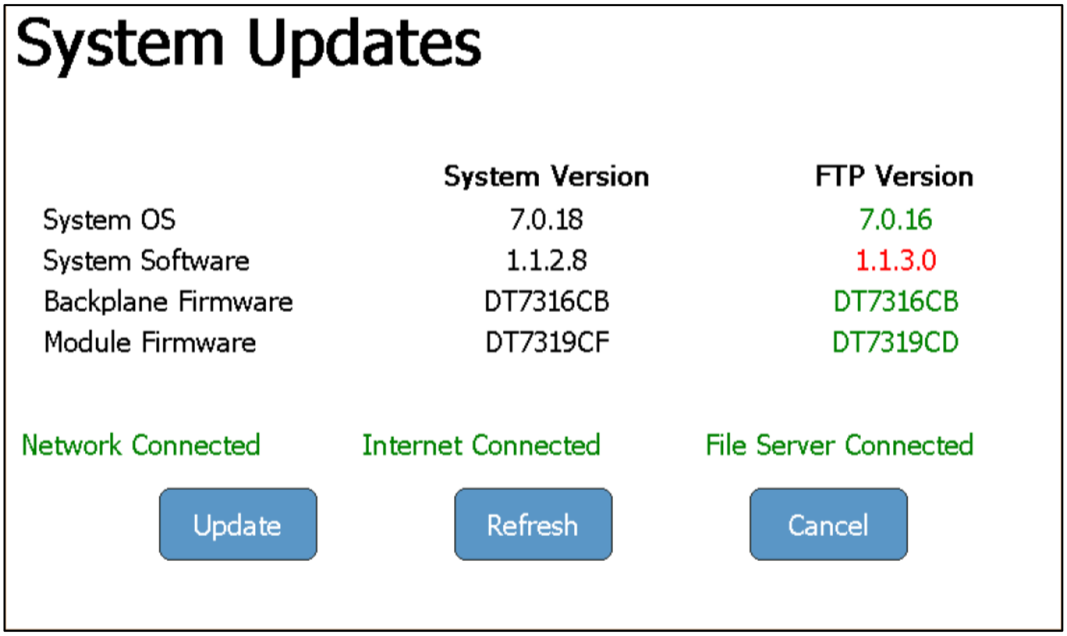
Press the **System Updates** key to display the following screen:



This screen allows the user to update the IPA-3400 and installed Flow Modules over the internet (FTP), or from a USB flash drive containing the update files. This screen also provides a function to restore to previous software versions using the **System Restore** key.

FTP

If the IPA-3400 is connected to an internet-connected network press the **FTP** key to display the following screen:

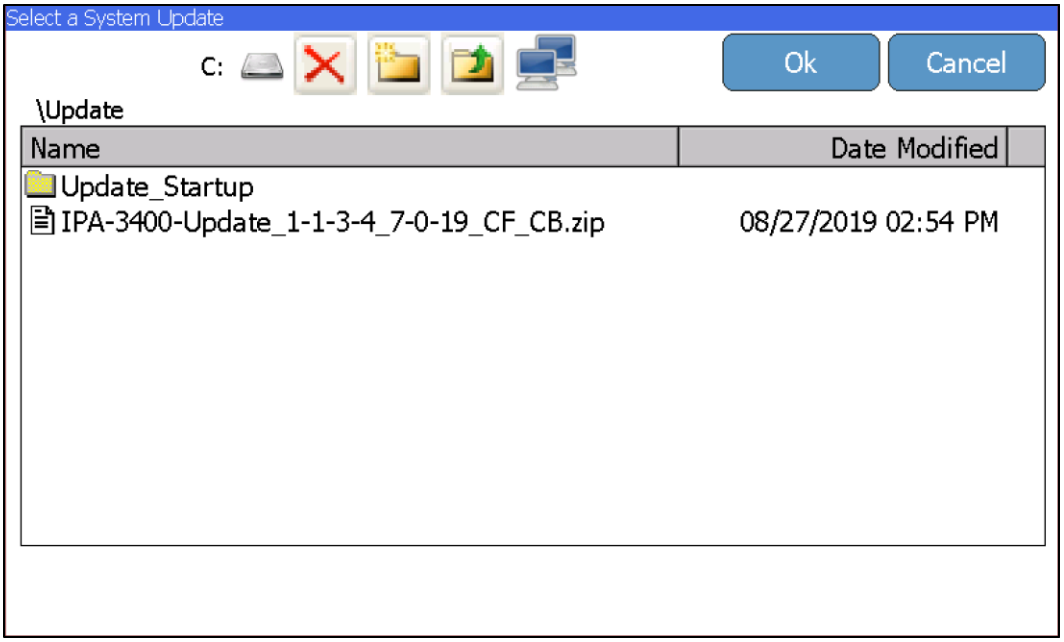


The current System Versions will be displayed. If the Network, Internet, and File Server indicate "connected" (as shown above) the available FTP Versions will also be displayed. If a newer version is available, it will be highlighted in a red color font. Press the **Update** key to begin the available updates.

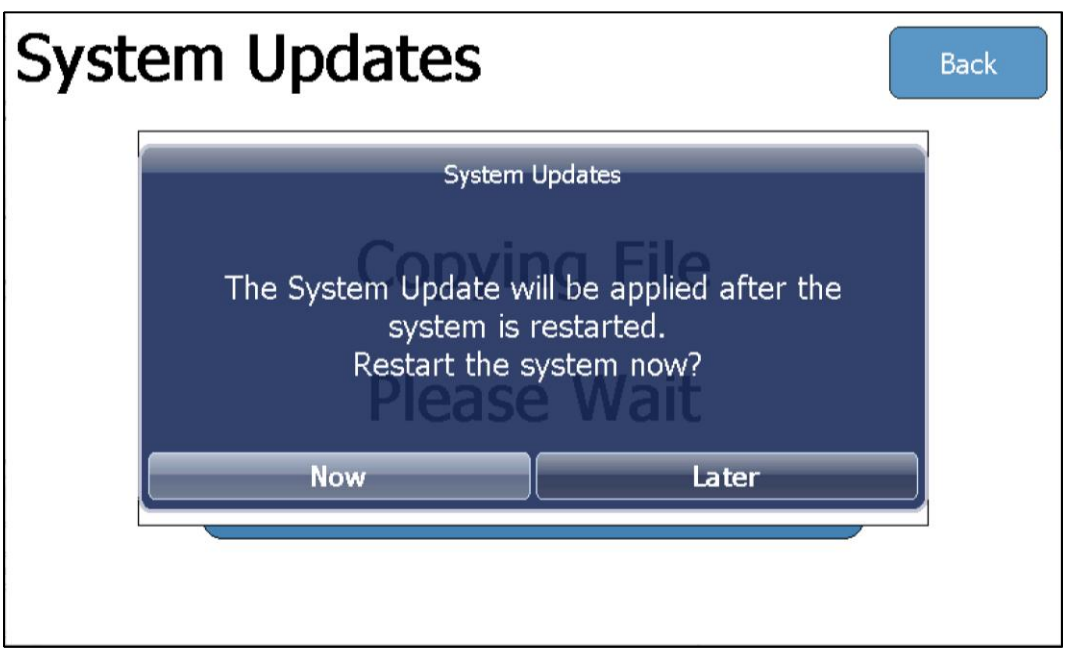
USB

Using a PC, download the latest software version .zip file from our website (www.bcgroupstore.com) and save the .zip file to a USB flash drive. Insert that USB flash drive in an available USB receptacle on the back of the IPA-3400.

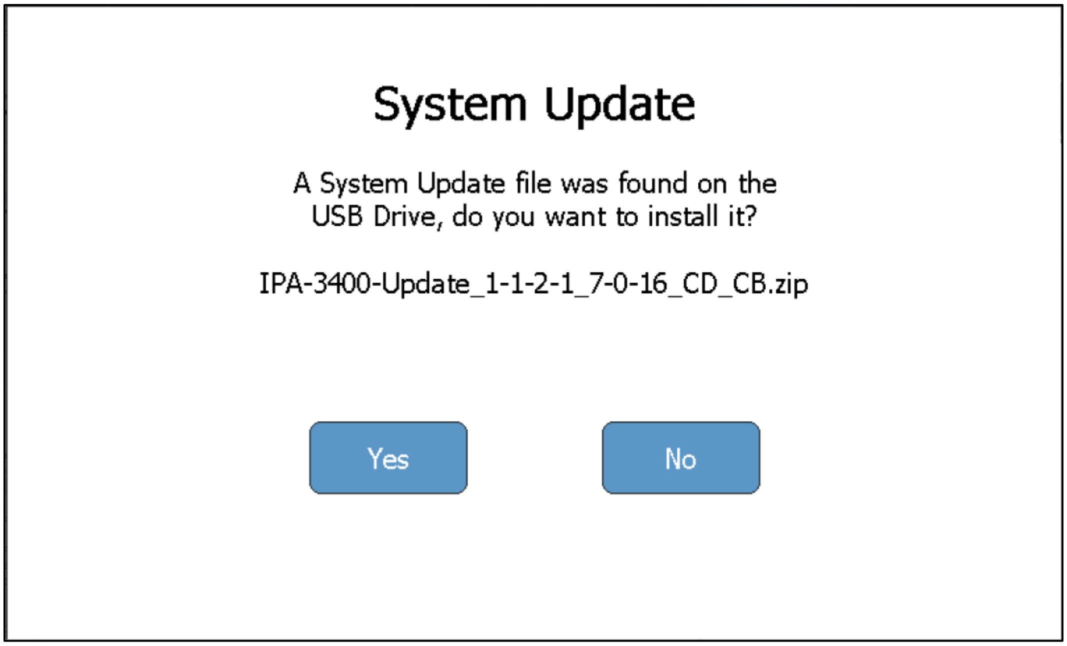
To update from a USB flash drive, press the USB key to display the following screen:



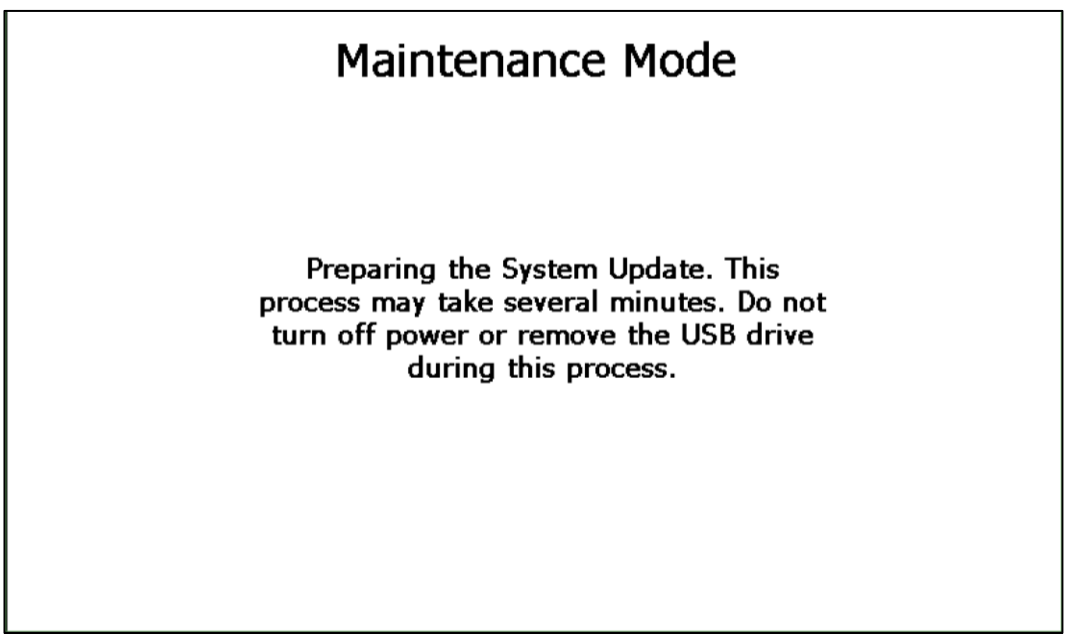
Browse to the update zip file on your USB drive and select OK. Please wait while the IPA-3400 copies the update zip file to the system hard drive. You will then be prompted to restart the system and install the update.

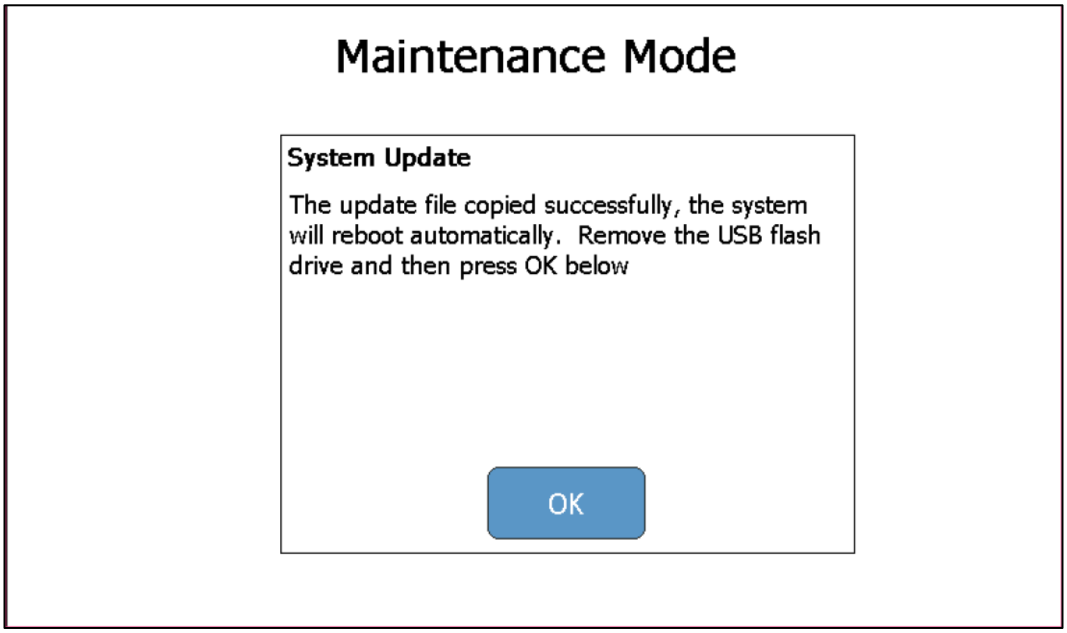


Additionally, you can place the update zip file in the root directory of a USB drive, connect the drive to the IPA-3400 while power is turned off, and then turn power on. The IPA-3400 will find the update zip on your USB drive and prompt you to install the update.

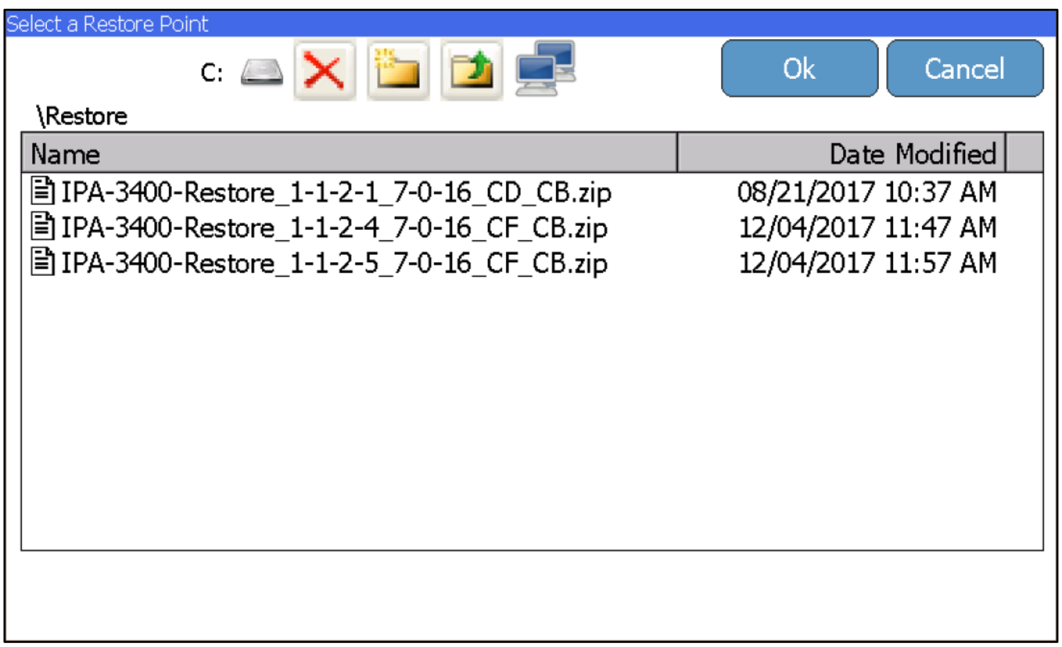


The IPA-3400 will then enter Maintenance Mode. Follow the on screen instructions.





System Restore
Each time the IPA-3400 software is updated, a backup of the previous version is created. If needed, press the **System Restore** key to display the following screen:



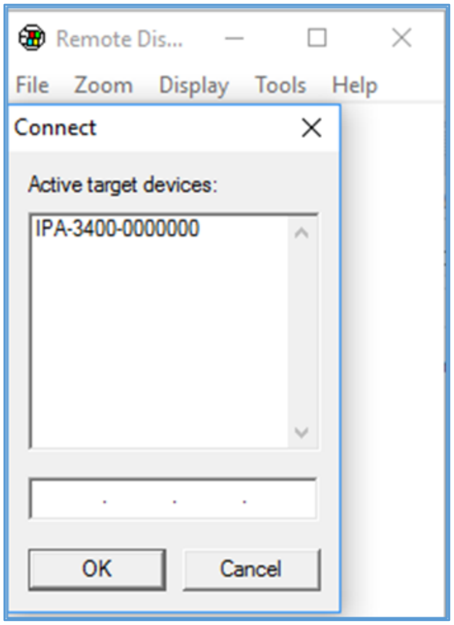
To restore a previous version, select the file and press the **Ok** key.

Remote Mode

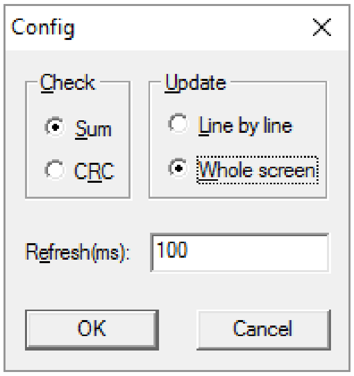
This mode allows the IPA-3400 to be controlled from a remote PC on a network. To activate Remote Mode, connect the IPA-3400 to the network then press the **Remote Mode** key.

Note: No visible changes will occur on the IPA-3400. A background application has been launched to allow the remote access.

Using a PC (on the same network as the IPA-3400) browse to the device in windows explorer by typing "\\IPA3400_xxxx\Utility" in the address field, where xxxx is the last four digits of the serial number of the IPA-3400. Copy the CERHost utility (cerhost.exe) file to the PC. Launch the CERHost file on the PC and select File – Connect. A window similar to the following will appear:

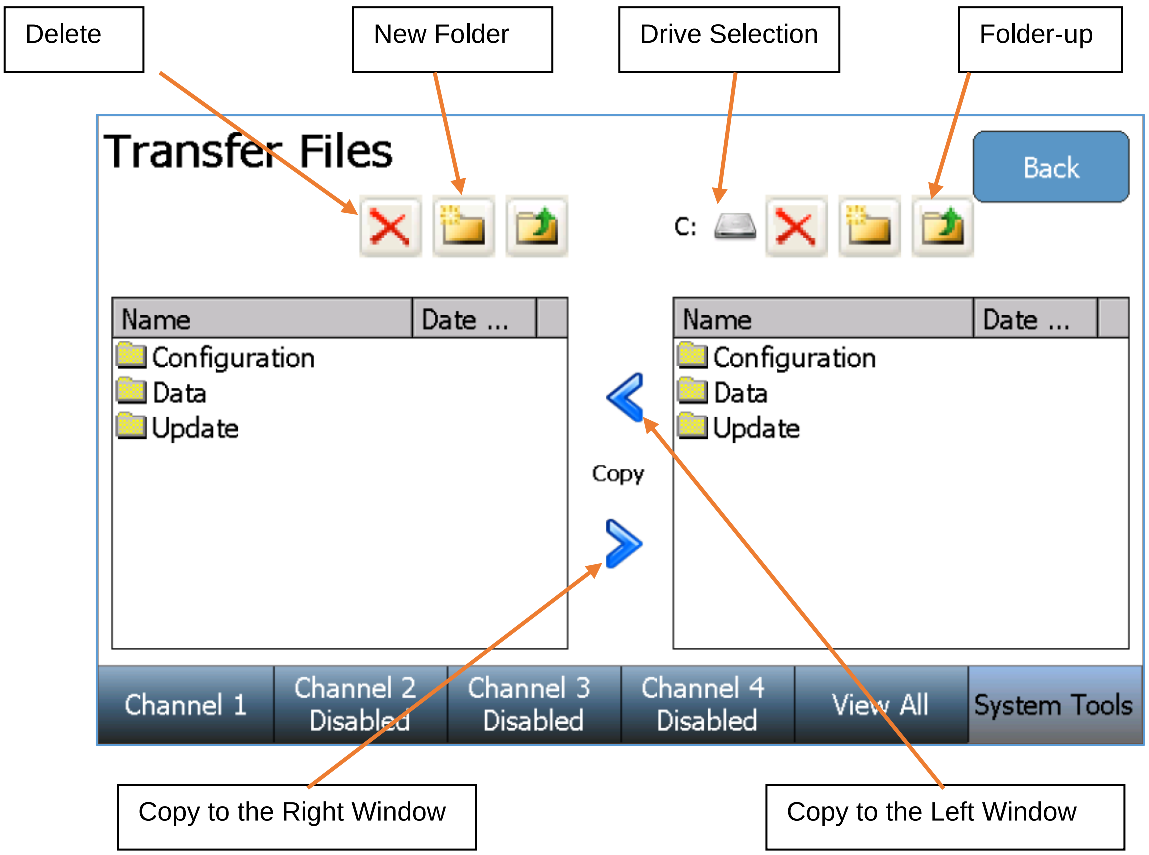


Select the listed IPA-3400 "Active target device" and select the OK button. The PC display may flicker or show lines. To fix this, select the CERHost application Tools - Config menu and set the Update selection to "Whole screen":



Transfer Files

The **Transfer Files** key allows the user to copy any folder or file between the IPA-3400 and an external USB flash drive, directly from the IPA-3400.



The left window is the IPA-3400 hard drive (C: drive). The right window allows the user to select the IPA-3400 (C: drive) or any inserted USB flash drive (D:, E:, F:, or G:) using the drive selection button. Highlight the folder or file to copy from either window, then press the copy arrow pointing towards the desired file location.

Networking

Press the **Networking** key to display the current network information:

Address Type – The user can select whether the IP address is received automatically from a DHCP server or configured for a Static IP address. **NOTE: After changing the Address Type, the system will need to be rebooted for the changes to take effect.**

IP Address – This field identifies the IP address when DHCP is enabled or allows the user to enter an IP address when the Static Address Type is selected.

Subnet Mask– This field identifies the Subnet Mask when DHCP is enabled or allows the user to enter a Subnet Mask when the Static Address Type is selected.

Default Gateway – This field identifies the Default Gateway when DHCP is enabled or allows the user to enter a Gateway address when the Static Address Type is selected.

MAC Address – This read only field identifies the MAC address of the IPA-3400.

Network Name – This identifies the IPA-3400 on the network. The default name is IPA3400_xxxx where “xxxx” is the IPA-3400 serial number. Network names must be unique for each device on a network.

Using a PC (on the same network as the IPA-3400) browse to the device in windows explorer by typing “\\IPA3400_xxxx\” in the address field, where xxxx is the serial number of the IPA-3400.

Flow Module Replacement

The IPA-3400 System is unique in that it uses a patented (US Patent No. 10,100,828) user-replaceable Flow Module design. This means the system can run with one to four Flow Modules in any of the four Module slots.

The IPA-3900-FM Flow Module is a self-contained unit that is a complete flow measuring assembly with all programming, model number, serial number, calibration data, and usage history stored onboard.

The Flow Modules are completely interchangeable within the unit and from unit to unit. The information in the Module is read by the IPA-3400 and displayed in the System Information screen in System Tools. This information is appended to all test data and is available on all test reports.

From a service or upgrade viewpoint, this provides for maximum flexibility. The user is in full control. If you buy a unit with just 1 Module, you can use this process to upgrade the system with up to three additional Modules.

IPA-3900-FM Flow Module

The individual Module is all that is required for annual calibration. Simply return the Module and continue to use the system. By staggering the calibration dates on the Modules, or having a spare Module, system down time is at a minimum.

Flow Modules can be replaced without any tools and do not require that any wiring or fluid connection be removed or installed. See the following page for Flow Module Installation details.

Flow Module Installation

NOTE: All IPA-3400 screws are self-retaining.
They can be loosened, but not removed.

- Turn the IPA-3400 Power Switch to the Off (O) position.

- Lift the IPA-3400 screen to its highest position using the tab at the bottom-right of the screen.

- Loosen the two IPA-3400 Access Screws and remove the IPA-3400 Flow Module Access Panel.

- Loosen the Module Screw on the bottom of the IPA-3400 that corresponds to the desired location.

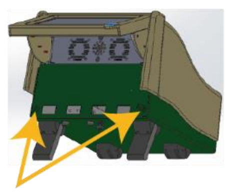
- Slide the Module Plug, or previous Flow Module out of the IPA-3400 chassis.

- Gently slide the new Flow Module into the IPA-3400 chassis.

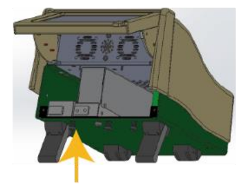
- Tighten the Module Screw under the IPA-3400 that corresponds to the installed Flow Module.

- Install the IPA-3400 Access Panel and gently tighten the two IPA-3400 Access Screws.

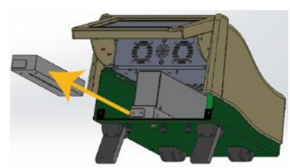
- Turn the IPA-3400 Power Switch to the On (I) position to begin using the new Flow Module.



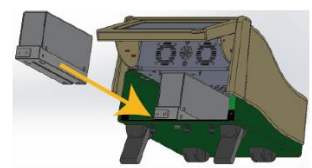
1) LOOSEN ACCESS PANEL SCREWS



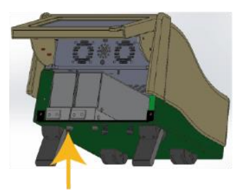
2) LOOSEN MODULE SCREW



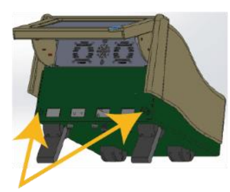
3) REMOVE MODULE PLUG



4) INSERT NEW MODULE



5) TIGHTEN MODULE SCREW



6) TIGHTEN ACCESS PANEL SCREWS

COMMUNICATION PROTOCOL

The communication protocol provides a means to completely configure and use the IPA-3400 from a PC. All of the functions available through the touch screen can be performed through the communication ports. All measurements made by the IPA-3400 are accessible as well. This provides for hands free or automated operation of the IPA-3400.

Communication Ports

The IPA-3400 has four USB ports that can be used to connect to a PC. The connection must be made using an FTDI USB null modem cable (BC Part number BC20-41360). This cable allows the PC to see the connection to the IPA-3400 as a serial port. To determine the serial port number, refer to the Ports section of the Device Manager on the PC. The Serial port is configured as 115,200 Baud Rate, 8 Data Bits, 1 Stop Bit, and No Parity.

Command Syntax

The command description is broken into 5 columns; the CHANNEL, the KEYWORD, the NODE, the SUBNODE, and the VALUE.

Keyword

The KEYWORD column provides the name of the command. The actual name of the command consists of one or more keywords since SCPI commands are based on a hierarchical structure, also known as a **tree system**.

In such a system, associated commands are grouped together under a common node in the hierarchy, analogous to the way leaves at a same level are connected at a common branch. This and similar branches are connected to fewer and thicker branches, until they meet at the root of the tree. The closer to the root, the higher a node is considered in the hierarchy. To activate a particular command, the full path to it must be specified.

This path is represented in the following tables by placing the highest node in the left-most position. Further nodes are indented one position to the right, below the parent node.

The highest level node of a command is called the Keyword, followed by the Node, Subnode, and then the Value.

Not all commands require the complexity of the full command path. For example, the “SYStem:PRESSureunit PSI” command doesn’t have a Node or Subnode.

Some commands allow for reading and writing data and some commands are Read Only. To indicate a read function, a question mark (?) is placed at the end of the command path. For example, a write command to change the Flow Rate for a Flow Test on Channel 1 to 100 mL/hr would be “CH1:CONFigure:FLOW:FLOWrate 100<cr>”, where <cr> indicates a carriage-return. For example, a Volume read command on Channel 3 would be “CH3:READ:VOLume?<cr>”, which would return a value of “xxx.x<cr><lf>” where <cr> is a carriage-return and <lf> is a linefeed.

Lowercase letters indicate the **long-form** of the command (for example, **CH1:CONFigure:OCCLusion:PARameter?**) and can be omitted for simplification. Uppercase letters indicate the abbreviated, or **short-form,** of the commands and must be included (for example, **CH1:CONF:OCC:PAR?**).

All commands sent to the unit are terminated with a Carriage Return.

NOTE: Commands can be entered in either upper or lowercase or a mixture of the two, uppercase and lowercase. Commands sent to the IPA-3400 are not case sensitive. Upper and lower cases are only used when documenting the commands.

Parameter Form

The PARAMETER FORM column indicates the number and order of parameters in a command and their legal values. Parameter formats are listed in angle brackets <> while string parameters are simply listed.

The vertical bar | can be read as “or” and is used to separate alternative parameter options.

The query form of a command is generated by appending a question mark “?” to the last keyword. However, not all commands have a query form, and some commands exist only in the query form.

Channel	Keyword	Node	Subnode	Value
CH1, CH2, CH3, CH4	CONFigure	FLOW	FLOWrate	0-1600
			VTBI	0-9999
			DURation	0 - 431999
			PARameter	InstFlow | AvgFlow | Volume
			UPPERTOLerance	based on parameter
			LOWERTOLerance	
			TOLerancetype	PERcent | VALue
			BACKPRESSure	-3.867 - 11.602 PSI
			STARTCONDition	MANual | AUTOmatic
			ENDCONDition	DURation | VOLume | DURATIONANDVolume | DURATIONORVolume

<table>
<tr><td rowspan="39"></td><td rowspan="16"></td><td rowspan="11">PCA</td><td>BASALflowrate</td><td>0-1600</td></tr>
<tr><td>DURation</td><td>0-431999</td></tr>
<tr><td>PARameter</td><td>FLOW | VOLume</td></tr>
<tr><td>EXPectedvalue</td><td>based on parameter</td></tr>
<tr><td>TOLerancetype</td><td>PERcent | VALue</td></tr>
<tr><td>UPPERTOLerance</td><td rowspan="2">based on parameter</td></tr>
<tr><td>LOWERTOLerance</td></tr>
<tr><td>LOCKouttime</td><td>0-999</td></tr>
<tr><td>LOADingdose</td><td>0-9999</td></tr>
<tr><td>BACKPRESSure</td><td>-3.867 - 11.602 PSI</td></tr>
<tr><td>STARTCONDition</td><td>MANual | AUTOmatic</td></tr>
<tr><td rowspan="4">OCClusion</td><td>PUMPtype</td><td>MANual</td></tr>
<tr><td>PARameter</td><td>PRESSure | NONE</td></tr>
<tr><td>UPPERlimit</td><td rowspan="2">based on parameter</td></tr>
<tr><td>LOWERlimit</td></tr>
<tr><td>MODE</td><td>FLOW | PCA | OCClusion</td><td></td></tr>
<tr><td rowspan="20">READ ?
Query only</td><td>SEConds</td><td colspan="2" rowspan="11"></td></tr>
<tr><td>PRESSure</td></tr>
<tr><td>VOLume</td></tr>
<tr><td>AVGflow</td></tr>
<tr><td>INSTflow</td></tr>
<tr><td>PERcenterror</td></tr>
<tr><td>PEAKSeconds</td></tr>
<tr><td>PEAKPressure</td></tr>
<tr><td>ALARMSeconds</td></tr>
<tr><td>MAXInst</td></tr>
<tr><td>MINInst</td></tr>
<tr><td rowspan="9">BOLUS</td><td>AVGFlow</td><td rowspan="5"></td></tr>
<tr><td>AVGVolume</td></tr>
<tr><td>AVGDuration</td></tr>
<tr><td>AVGInterval</td></tr>
<tr><td>COUnt</td></tr>
<tr><td rowspan="4">X = 0,1,2,3,etc</td><td>AVGFlow</td></tr>
<tr><td>VOLume</td></tr>
<tr><td>DURation</td></tr>
<tr><td>INTerval</td></tr>
<tr><td rowspan="4">KEY</td><td>SELect</td><td colspan="2">Selects that Channel in the interface</td></tr>
<tr><td>START</td><td colspan="2">Starts test in current test mode on specified Channel</td></tr>
<tr><td>STOP</td><td colspan="2">Stops test in current test mode on specified Channel</td></tr>
<tr><td></td><td>ALARM</td><td colspan="2">Triggers the Alarm button in an Occlusion test.</td></tr>
<tr><td>ALL</td><td>KEY</td><td>SELect</td><td colspan="2">Selects View All in the interface</td></tr>
<tr><td>SYStem</td><td>PRESSureunit</td><td>mmHG | PSI |
mBar | kPa</td><td colspan="2"></td></tr>
</table>

<table>
<tr><td rowspan="7"></td><td>AUDiovolume</td><td>HIGH | MEDIUM
| LOW | SILent</td><td colspan="2"></td></tr>
<tr><td>SAMplingwindow</td><td>1 - 10</td><td colspan="2"></td></tr>
<tr><td rowspan="2">TIMEanddate</td><td>TIME</td><td>hh:mm (24 hour format)</td><td></td></tr>
<tr><td>DATE</td><td>mm/dd/yyyy</td><td></td></tr>
<tr><td rowspan="2">NETwork
Query Only</td><td>IP</td><td>returns IP address</td><td rowspan="2"></td></tr>
<tr><td>NETworkname</td><td>returns Network Name</td></tr>
</table>

MANUAL REVISIONS

Revision #	**Revisions Made**
Rev 01	Preliminary Manual
Rev 02	Update for clarity and user-selectable Language support.
Rev 03	New Software v.2.0
Rev 04	Update to Specifications

LIMITED WARRANTY

WARRANTY: BC GROUP INTERNATIONAL, INC. WARRANTS ITS NEW PRODUCTS TO BE FREE FROM DEFECTS IN MATERIALS AND WORKMANSHIP UNDER THE SERVICE FOR WHICH THEY ARE INTENDED. THIS WARRANTY IS EFFECTIVE FOR TWELVE MONTHS FROM THE DATE OF SHIPMENT.

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BC GROUP INTERNATIONAL, INC. IS NOT LIABLE FOR ANY INCIDENTAL OR CONSEQUENTIAL DAMAGES.

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REMEDIES: THE PURCHASER'S SOLE AND EXCLUSIVE REMEDY SHALL BE: (1) THE REPAIR OR REPLACEMENT OF DEFECTIVE PARTS OR PRODUCTS, WITHOUT CHARGE. (2) AT THE OPTION OF **BC GROUP INTERNATIONAL, INC.**, THE REFUND OF THE PURCHASE PRICE.

SPECIFICATIONS

FLOW MEASUREMENT		
RANGE	0.01 ml/hr to 2600 ml/hr	
FLOW RATE	0.10 ml/hr to 1600 ml/hr	
FLOW RESOLUTION	0.010 ml/hr (10 µl/hr)	
RESOLUTION	10 µl/hr	
ACCURACY (After 30 sec)	±1% reading + 1LSD	0.10 to 9.90 ml/hr
	±1% reading	10 to 700 ml/hr
	±2% reading	700 to 1600 ml/hr
BACK-PRESSURE CONTROL	RESOLUTION	0.001 PSI (0.05 mmHg)
	RANGE	-3.867 to 11.602 PSI (-200 to 600 mmHg)
	ACCURACY	0.1% FS
MIN. VOLUME	0.05 ml (50 µl)	
MEASUREMENT RATE	1 Hz	
CHANNELS	1, 2, 3 or 4 (user-installable)	

VOLUME MEASUREMENT	
RANGE	0 to 9999 ml
RESOLUTION	0.001 mL (1.0 µl)
ACCURACY	±1% Reading ± 20 µL, after 100 µL

PRESSURE MEASUREMENT	
RANGE	-5 to 50 PSI (-258.57 to 2585.75 mmHg)
RESOLUTION	0.001 PSI (0.05 mmHg)
ACCURACY	±0.1% FS
ENGINEERING UNITS	PSI (default), mmHg, Bar, kPa
MEASUREMENT RATE	1 Hz

PCA/BOLUS MEASUREMENT	
DISPLAY RANGE	0.1 to 100 ml
MEASUREMENT RANGE	0.5 to 100 ml
ACCURACY	±1%
MIN. BOLUS VOLUME	0.01 ml (10 µl)

ELAPSED TIME	
RANGE	0-120 hours
RESOLUTION	1 second
ACCURACY	0.5 second

OCCLUSION (PRESSURE) TEST	
RANGE	-258.57 to 2585.75 mmHg (-5 to 50 PSI)
RESOLUTION	0.05 mmHg (0.001 PSI)
ACCURACY	±0.1% FS

ELECTRICAL		
EXTERNAL CONNECTIONS	USB HOST PORTS	4 USB-A
	SUPPORTED DEVICES	HID Keyboard HID Mouse HID Barcode Scanner FAT 32 Formatted Flash Drive
	ETHERNET	10/100/1000 LAN
AUX	Ports	4 (1 per Channel)
POWER	Input Voltage	90-264 VAC
	Input Frequency	50-60 Hz
	Power Consumption	90 VA
	Fuses	5 x 20 mm, T2 4A

PHYSICAL & ENVIRONMENTAL		
MEDIA	Distilled or Deionized Water	
MEDIA CONNECTIONS (FLUID FITTINGS)	INLET	Female Luer
	DRAIN	Male Luer Lock
DISPLAY	7” Widescreen Color Touchscreen, 800 x 480 px	
MEMORY	CALIBRATION	EEPROM
	DATA RECORDS	INTERNAL 32 GB Flash memory
CONSTRUCTION	ENCLOSURE	PC/ABS Plastic, Aluminum
SIZE (H x W x D)	7.8 x 9.1 x 10.2 Inches (mm)	
WEIGHT	IPA-3400-1	< 8 Lbs (3.63 kg)
	IPA-3400-2	< 10 Lbs (4.54 kg)
	IPA-3400-3	< 12 Lbs (5.44 kg)
	IPA-3400-4	< 14 Lbs (6.35 kg)
OPERATING RANGE	15 to 40 °C (59 to 86 °F), 10 to 80% RH, Non-Condensing	
STORAGE RANGE	0 to 50 °C (32 to 122 °F)	
DATA STORAGE	Internal 32 GB	

NOTES

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IPA-3400 Series User Manual
09/21 – Rev 04

Made in the USA